PEARSON EDEXCEL INTERNATIONAL A LEVEL PHYSICS · WPH12

Unit 2: Waves and Electricity

Revision notes, spec point by spec point (Waves and the Particle Nature of Light; Electric Circuits)

How to use these notes. Every section starts with a spec-point badge and a short summary of what that spec point asks, so you always know which point you are revising. The header at the top of each page names the spec point(s) on that page.

BEYOND THE SPEC, BUT EXAMINED

Content the specification does not spell out, but that past-paper mark schemes have rewarded.

EXAM TIP

How to pick up the marks.

MARK-SCHEME LANGUAGE

Words and phrases mark schemes credit.

COMMON MISTAKE

What examiners keep seeing go wrong.

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Topic: Waves and the Particle Nature of Light

SPEC 33

Amplitude, frequency, period, speed and wavelength

WHAT THIS SPEC POINT ASKS

Know what amplitude, frequency, period, wave speed and wavelength mean.

33 What a progressive wave is

- A **progressive (travelling) wave** transfers **energy** from one place to another **without transferring matter**. The particles of the medium (or the field values, for light) just **oscillate about a fixed equilibrium position**.
- Mechanical waves (sound, waves on a string) need a medium. Electromagnetic waves don't: they cross a vacuum.

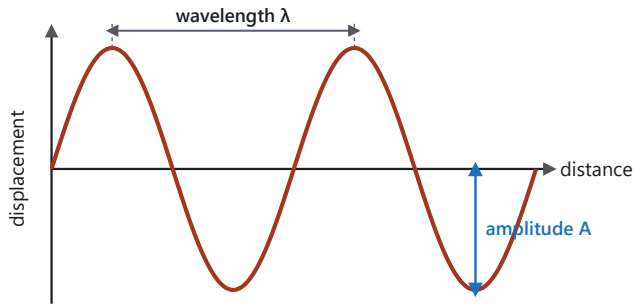
33 The five terms

Term	Symbol, unit	Meaning
Displacement	x or s, m	distance and direction of a point from its equilibrium position
Amplitude	A, m	the **maximum displacement** from equilibrium (not peak to trough)
Wavelength	λ, m	the distance between two **adjacent points in phase**, e.g. crest to crest
Period	T, s	the time for **one complete oscillation** of a point on the wave
Frequency	f, Hz	the **number of oscillations per second** (1 Hz = 1 s^{-1})
Wave speed	v, m s^{-1}	the distance the wave (the energy) travels per second

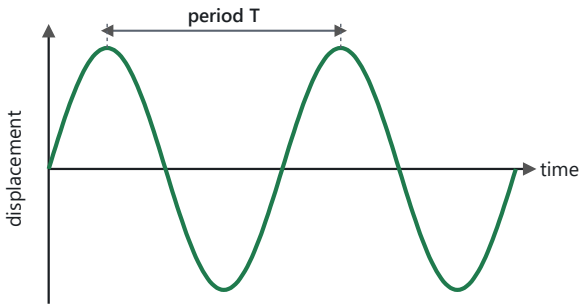
Frequency and period are reciprocals:

$$f = 1 / T$$

33 Which graph gives which quantity



Displacement–distance: a snapshot of the whole wave



Displacement–time: one point on the wave

Same shape, different x-axis. Read λ off a distance axis and T off a time axis. Both show the amplitude.

Graph	Shows	Read off it
Displacement–distance	a snapshot of the whole wave at one instant	**wavelength** λ and amplitude A
Displacement–time	how **one point** moves as time passes	**period** T (so f = 1/T) and amplitude A

To find the speed you need λ from the first graph and f from the second. You can never get λ from a displacement–time graph.

33 Reading an oscilloscope

- Period = (number of horizontal divisions for one cycle) × (time-base setting).
- Measure across **several complete cycles** and divide. This cuts the percentage uncertainty.

WORKED EXAMPLE: FREQUENCY FROM AN OSCILLOSCOPE TRACE

Three complete cycles span 7.5 divisions. The time base is 2.0 ms div^{-1}.
Time for 3 cycles = 7.5 × 2.0 ms = 15 ms, so T = 15 ms ÷ 3 = 5.0 ms = 5.0×10^{-3} s.
f = 1/T = 1 ÷ (5.0×10^{-3} s) = **200 Hz** (2 s.f.).

MARK-SCHEME LANGUAGE

- "Use of f = 1/T"; "Use of v = fλ".
- Oscilloscope questions: "uses at least 3 waves to determine T" is a marking point on its own (WPH12 Jun 2024).
- Amplitude: "maximum displacement" from the equilibrium/rest position.

Often asked as: MCQs on which graph gives which quantity, reading T from a trace, comparing two traces on one screen.

COMMON MISTAKE

Reading a "wavelength" off a displacement–time graph. The distance between peaks there is the **period**. Measuring amplitude peak to trough (that is 2A).
Leaving the time base in ms (or μs) when working out f. Convert to seconds first.

BEYOND THE SPEC, BUT EXAMINED

Two sound traces on one screen, same time base: if wave 1 shows one-third as many cycles, its frequency is a third, so its wavelength is three times as long, because the speed of sound is the same for both and v = fλ (WPH12 Jun 2024).

SPEC 34 The wave equation v = fλ

WHAT THIS SPEC POINT ASKS

Use v = fλ to link wave speed, frequency and wavelength.

34 Where it comes from

In one period T the source makes one complete oscillation, and the wave moves forward one wavelength. Speed = distance ÷ time:

$$v = \lambda / T = f\lambda$$

v = wave speed (m s^{-1}), f = frequency (Hz), λ = wavelength (m). It is on the formulae sheet.

34 What sets what

- The **source** sets the frequency. The **medium** sets the speed. The wavelength then follows: λ = v/f.
- At a fixed speed, f and λ are inversely proportional: double f and λ halves.

- When a wave crosses into a new medium its **frequency stays the same**. Its speed changes, so its wavelength changes in proportion (light slows in glass, so λ gets shorter).

Wave	Typical speed
Sound in air (room temperature)	about 330–340 m s^{-1}
Sound in water	about 1500 m s^{-1}
Sound in steel	about 5000–6000 m s^{-1}
Any EM wave in a vacuum (and almost in air)	$c = 3.00 \times 10^8$ m s^{-1}

WORKED EXAMPLE: V = FΛ, AND A CHANGE OF MEDIUM

(a) A 512 Hz tuning fork in air, v = 340 m s^{-1}: $\lambda = v/f = 340 \div 512 =$ **0.664 m** (3 s.f.).
(b) Light of frequency 5.00×10^{14} Hz goes from air into water, where its speed is 2.26×10^8 m s^{-1}.
In air: $\lambda = 3.00 \times 10^8 \div 5.00 \times 10^{14} = 6.00 \times 10^{-7}$ m (600 nm).
In water: f is unchanged, so $\lambda = 2.26 \times 10^8 \div 5.00 \times 10^{14} = 4.52 \times 10^{-7}$ m = **452 nm** (3 s.f.).

MARK-SCHEME LANGUAGE

- "Use of v = fλ" is almost always the first mark, credited even if only implied by the numbers.
- In graph questions, a plot of λ against 1/f is a straight line through the origin and "gradient = v" must be stated.

Often asked as: one-step calculations with prefixes (kHz, MHz, nm), λ found from a stationary-wave pattern first, comparing travel times of light and sound.

COMMON MISTAKE

Thinking the frequency changes when a wave enters a new medium. It doesn't: v and λ change together (WPH12 Jan 2020 examiner report).
Prefix slips: 1 nm = 10^{-9} m, 1 μs = 10^{-6} s, 1 MHz = 10^6 Hz.

BEYOND THE SPEC, BUT EXAMINED

Unfamiliar wave-speed formulas given in the question, which you just substitute into: deep-water waves $v = \sqrt{(g\lambda/2\pi)}$ and shallow-water waves $v = \sqrt{(gd)}$ (WPH12 Jan 2020).
Seismic S-waves have $v = \sqrt{(G/\rho)}$. A liquid has G = 0, so S-waves can't travel through liquids (WPH12 Jun 2022).

SPEC 35

Longitudinal waves

WHAT THIS SPEC POINT ASKS

Describe a longitudinal wave using the displacement of the molecules and the variation in pressure.

35 What it is

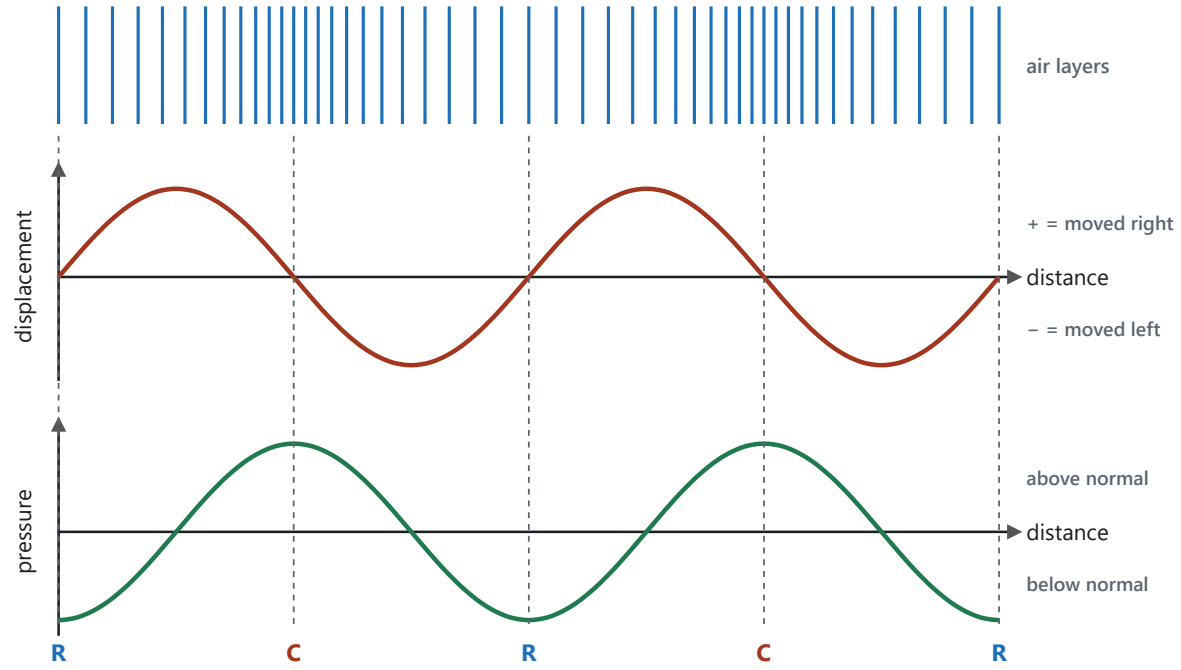
- In a **longitudinal** wave the particles oscillate **parallel to the direction the wave travels** (the direction of energy transfer).
- Examples: **sound**, ultrasound, seismic P-waves, a slinky pushed back and forth along its length.
- They need a medium, and they **cannot be polarised**.

35 Compressions and rarefactions

- A **compression** is a region where the molecules are pushed **closer together**, so the **pressure is above normal** (a pressure maximum).
- A **rarefaction** is a region where the molecules are **further apart**, so the **pressure is below normal** (a pressure minimum).
- Each layer of air moves forwards and backwards about its own rest position. No air travels with the wave.
- One wavelength = the distance from one compression to the next (or one rarefaction to the next).

35 Displacement and pressure are a quarter-cycle apart

Look at one compression. The air on its left has moved **right** (towards it) and the air on its right has moved **left** (towards it). The layer right at the centre has been squeezed from both sides but **has not moved**: its displacement is **zero**. The same is true at the centre of a rarefaction (neighbours have moved away from it). Maximum displacement is halfway between a compression and a rarefaction, where the pressure is normal.



C = compression (pressure maximum), R = rarefaction (pressure minimum).
Displacement is zero at the centre of every C and every R.

The layers bunch at C and spread at R. Displacement is zero at every C and every R; pressure is greatest at C and least at R. The two graphs are a quarter of a wavelength apart.

MARK-SCHEME LANGUAGE

- Longitudinal: "oscillations/vibrations are **parallel** to the direction of wave travel / energy transfer / propagation".
- "Zero displacement at **all** compressions **and** rarefactions."
- Rarefactions are "points where the **pressure** is a minimum" (not where displacement is a minimum).

Often asked as: MCQs on where displacement is zero; sketch the displacement–distance graph from a pressure–distance graph; define and compare transverse and longitudinal.

COMMON MISTAKE

Thinking only the centre of a compression has zero displacement. The centre of every rarefaction does too (WPH12 Oct 2022 examiner report).
Assuming a sine-shaped graph means the wave is transverse. A displacement–distance graph of sound is a graph, not a picture.
Writing "the particles move along with the wave". They oscillate about fixed positions.

BEYOND THE SPEC, BUT EXAMINED

Comparing a sound wave with a stationary wave on a string (5 marks): same frequency; different speeds and wavelengths; every point of the sound wave has the same amplitude but the stationary wave's amplitude varies; the sound wave transfers energy, the stationary wave doesn't; string transverse, sound longitudinal. Don't contradict yourself ("different speeds but the same wavelength" can't be true at the same frequency) (WPH12 Jun 2023).

SPEC 36 Transverse waves

WHAT THIS SPEC POINT ASKS

Describe transverse waves.

36 What it is

- In a **transverse** wave the oscillations are **perpendicular to the direction of wave travel** (the direction of energy transfer).
- Highest points are **crests (peaks)**, lowest points are **troughs**.
- Examples: **all electromagnetic waves** (light, radio, X-rays), waves on a string or rope, seismic S-waves, a slinky shaken side to side.
- Transverse waves **can be polarised** (49). That is the one behaviour that separates them from longitudinal waves.

	Transverse	Longitudinal
Oscillation direction	perpendicular to travel	parallel to travel
Features	crests and troughs	compressions and rarefactions
Examples	EM waves, string, S-waves	sound, ultrasound, P-waves
Polarisation	yes	**no**
Reflection, refraction, diffraction, superposition, interference	yes	yes

36 Which way does a point move next?

On a displacement–distance snapshot of a transverse wave moving to the right:

1. Redraw the wave shifted a tiny distance to the right.
2. See whether the point's new displacement is higher or lower. That is the way it moves now.
3. Points at a crest or trough are momentarily **stationary**; points at zero displacement move **fastest**.

MARK-SCHEME LANGUAGE

- "Oscillations/vibrations **perpendicular** to the direction of wave travel / energy transfer / propagation" (1 mark).
- "Only transverse waves can be polarised." Refraction, diffraction and superposition happen for both kinds.

Often asked as: 1-mark definitions, transverse v longitudinal (2 marks), MCQs on which property needs a transverse wave.

COMMON MISTAKE

"Perpendicular to the wave" on its own is vague. Say perpendicular to the **direction of travel / energy transfer**.
Saying all transverse waves travel at the same speed. Only EM waves share a speed (in a vacuum).

BEYOND THE SPEC, BUT EXAMINED

Radio waves from an aerial are plane polarised, so a receiving aerial must be aligned with the transmitting one, acting like a polarising filter (WPH12 Jan 2026).

SPEC 37

Graphs of transverse, longitudinal and stationary waves

WHAT THIS SPEC POINT ASKS

Draw and interpret graphs of transverse and longitudinal waves, including stationary waves.

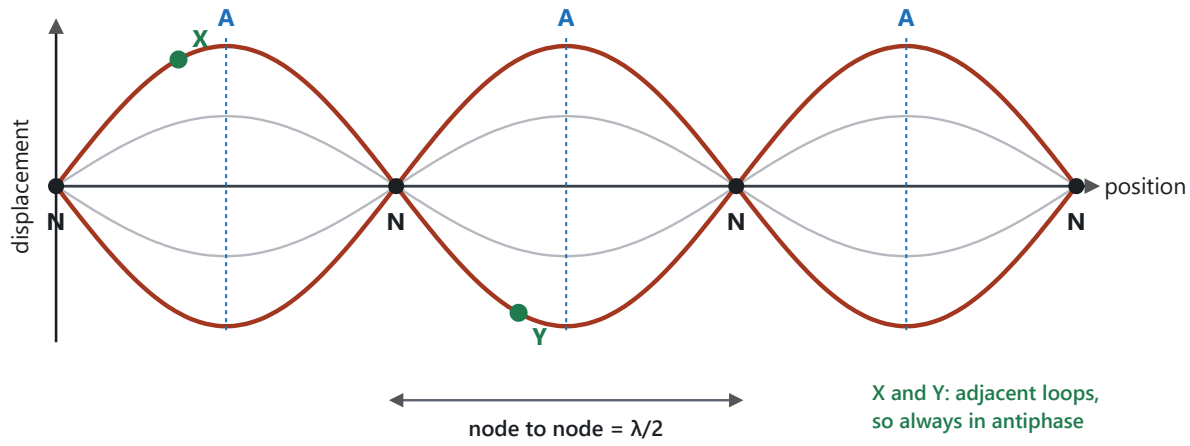
37 The checklist for any wave graph

1. **Read the axes first.** Distance on the x-axis → λ; time on the x-axis → T.
2. **Displacement** on the y-axis gives the amplitude. **Pressure** on the y-axis (sound only) shows compressions as maxima.
3. For a longitudinal wave, + displacement means "moved in the direction of travel", − means "moved back". The graph is not a picture of the wave.

37 Longitudinal: converting between the two graphs

- Pressure maxima (compressions) and minima (rarefactions) are where the displacement graph crosses **zero**.
- So the displacement graph is the pressure graph shifted by **a quarter of a wavelength** (see the figure in 35).
- When sketching: mark every C and R, put a zero of displacement at each, and draw a full sine curve through them with the same wavelength.

37 Stationary waves on a graph



Snapshots of a stationary wave. The red curves are the extreme positions; the grey ones are in-between instants. Nodes (N) never move. X and Y are in adjacent loops.

- The amplitude **depends on position**: zero at nodes, maximum at antinodes.
- Every point between two adjacent nodes moves **in phase** (they all reach their maximum together).
- Points in **adjacent loops** are in **antiphase** (180° apart): when one is up, the other is down.
- Twice per cycle the whole string is flat (the middle grey line), and every point is moving fastest at that moment.

WORKED EXAMPLE: READING A STATIONARY-WAVE SNAPSHOT

A string shows three loops over 1.80 m at 50 Hz.
Three loops = 3 × λ/2, so λ = 2 × 1.80 ÷ 3 = 1.20 m.
Points in the first and second loops: antiphase. Two points in the same loop: in phase, but with different amplitudes unless they are the same distance from a node.
Wave speed of the travelling waves that make it: v = fλ = 50 × 1.20 = **60 m s^{-1}** (2 s.f.).

MARK-SCHEME LANGUAGE

- Phase between two points: "in antiphase" because they are "in adjacent node-to-node regions/loops" (WPH12 Jun 2022).
- Different amplitudes: "S is at an antinode **and** T is between a node and an antinode".
- Drawing: correct shape, a node at each fixed end, antinode(s) in the right place.

Often asked as: phase and amplitude of points on a stationary wave (MCQ), sketch a pattern, read an oscilloscope trace, sketch displacement from pressure.

COMMON MISTAKE

Saying points in different loops are in phase. Adjacent loops are always in antiphase (WPH12 Jan 2022 examiner report).
Saying a node has "minimum displacement". It has **zero amplitude**: its displacement is always zero.

SPEC 38

Core practical 4: speed of sound in air

WHAT THIS SPEC POINT ASKS

Find the speed of sound in air using a two-beam oscilloscope, a signal generator, a loudspeaker and a microphone.

38 Principle

The oscilloscope shows two traces at once: the signal driving the loudspeaker and the signal picked up by the microphone. Moving the microphone **one wavelength** further from the speaker brings its trace back to the same phase relationship with the generator's trace. Measure λ that way, read f from the generator, and use v = fλ.

CORE PRACTICAL 4: METHOD IN BRIEF

1. Connect the signal generator to the loudspeaker and to channel 1 of the oscilloscope. Connect the microphone to channel 2. Set f to about 2–5 kHz.
2. Put the speaker and microphone on a bench along a metre rule. Move the microphone until the two traces are **in phase** (peaks line up). Record its position.
3. Slide the microphone away from the speaker, counting each time the traces come back **in phase**. Each return = one more wavelength. Move through **several** (e.g. 5) wavelengths and record the final position.
4. λ = total distance ÷ number of wavelengths.
5. Check f by measuring T on the oscilloscope (several cycles), f = 1/T, as well as reading the generator dial.
6. Repeat for several frequencies.

Analysis: plot λ (y) against 1/f (x). It is a straight line through the origin with **gradient = v**.

38 Variables

- **Independent:** frequency of the signal generator.
- **Dependent:** wavelength (microphone distance moved per cycle of phase).
- **Control:** room temperature (sound speed depends on it), speaker and microphone heights, keep the same "in phase" judgement each time.

38 Errors and improvements

- Judging exactly when the traces line up is the biggest uncertainty. Reduce it by measuring across **many wavelengths**, and by using lower frequencies (longer λ), so the same ± distance is a smaller percentage.
- **Reflections** from the bench, walls and ceiling add extra waves that blur the pattern. Work away from walls, put soft material on the bench.
- No reaction time is involved: the oscilloscope does the timing.
- Safety: keep the volume low.

WORKED EXAMPLE: PROCESSING CP4 RESULTS

At f = 4.00 kHz the traces are in phase at 12.0 cm and again, five wavelengths later, at 54.5 cm. Each reading is ±0.5 cm.
5λ = 54.5 − 12.0 = 42.5 cm, so λ = 8.50 cm = 0.0850 m.
$v = f\lambda = 4.00 \times 10^3 \times 0.0850 =$ **340 m s^{-1}** (3 s.f.).
Uncertainty in the distance = 0.5 + 0.5 = ±1.0 cm. Over five wavelengths: 1.0 ÷ 42.5 × 100 = 2.4 %. Measured over one wavelength (8.5 cm) it would be 12 %.

MARK-SCHEME LANGUAGE

- Method marks: microphone and speaker both on the oscilloscope; start with traces in phase (or antiphase); move the microphone until they are next in phase (antiphase); "distance moved = λ" (or λ/2 from in phase to antiphase); f from the generator or from T on the screen; then v = fλ.
- The final mark wants the **route**, e.g. "plot λ against 1/f, gradient = v" or "multiply λ by 1/T", not just "use v = fλ".
- The oscilloscope beats the ear: "hard for a person to judge when the sound is loudest/quietest"; "the oscilloscope can measure amplitude more accurately" (WPH12 Jun 2019).
- Percentage uncertainty is larger for a shorter λ (higher f). Don't write "uncertainty" when you mean "percentage uncertainty".

Often asked as: describe the method (5 marks), interpret two traces, % uncertainty at two frequencies, the two-speaker variant.

BEYOND THE SPEC, BUT EXAMINED

Two-speaker variant: the microphone finds quiet points where the path difference is λ/2. Path lengths come from Pythagoras, e.g. $\sqrt{(3.00^2 + 3.34^2)} - \sqrt{(3.00^2 + 1.66^2)} = 4.49 - 3.43 = 1.06$ m = λ/2, so λ = 2.12 m and v = 160 Hz × 2.12 m = 339 m s^{-1}. Reflections from walls and ceiling make the maxima and minima less clear (WPH12 Jun 2019).
Resonance-tube versions of the experiment: see 41.

SPEC 39 Wavefronts, coherence, path difference, superposition, interference and phase

WHAT THIS SPEC POINT ASKS

Know what wavefront, coherence, path difference, superposition, interference and phase mean.

39 The definitions

Term	Meaning
Phase	where a point is in its cycle of oscillation. **Phase difference** is how far one oscillation is ahead of or behind another, as a fraction of a cycle, given as an **angle** (degrees or radians)
Wavefront	a line (or surface) joining points on a wave that are **in phase**, e.g. along one crest
Path difference	the **difference in distance** travelled by two waves from their sources to the same point, in metres (or in wavelengths)
Superposition	when two or more waves meet, the resultant displacement at any point is the **sum of the individual displacements**
Coherence	two sources are coherent if they have the **same frequency** and a **constant phase difference**
Interference	the superposition of waves from **coherent** sources, giving a steady pattern of maxima and minima

39 Constructive and destructive interference

	Constructive	Destructive
Waves arrive	**in phase**	in **antiphase**
Phase difference	0, 360° (2π rad), …	180° (π rad), 540°, …
Path difference	**nλ** (n = 0, 1, 2, … a whole number)	**(n + ½)λ**: λ/2, 3λ/2, …
Result	larger amplitude: loud, bright	smaller amplitude: quiet, dark. Zero only if the two amplitudes are equal

39 Why coherence matters

- If the phase difference between two sources keeps changing, the maxima and minima move around faster than you can see or hear, so there is no steady pattern.
- **Two separate lamps** are not coherent: each atom emits light in short random bursts, so the phase relationship changes all the time (and the light contains many frequencies).
- **Coherent set-ups:** two loudspeakers driven by the **same** signal generator; one laser beam split by two slits or a grating.
- A laser is monochromatic (one frequency) and coherent.

39 How to write an interference explanation

1. Say the waves **superpose / interfere** where they meet.
2. Give the **path difference** at that point (nλ or (n + ½)λ, saying n is an integer).
3. Link it to **phase**: in phase or antiphase.
4. Name the result: constructive or destructive, so a maximum (loud/bright) or minimum (quiet/dark).

MARK-SCHEME LANGUAGE

- Coherent: "same frequency **and** constant phase difference / relationship". "Same phase" doesn't score (WPH12 Jan 2024 examiner report).
- Wavefront: "a line on which all points are in phase" (WPH12 Oct 2023).
- "Path difference = nλ **where n is an integer**": just "nλ" lost the mark for most candidates (WPH12 Oct 2024).
- "Phase difference must refer to an angle, path difference to a distance." "Out of phase" is not accepted for antiphase.

Often asked as: definitions (1 mark each), why two lamps aren't coherent, 6-mark interference explanations (two speakers, noise-cancelling headphones, beats).

COMMON MISTAKE

Mixing up phase and path difference: "a phase difference of λ/2" is wrong (it is π rad); "a path difference of 180°" is wrong (it is λ/2).
At the central point between two in-phase sources, the path difference is **zero**: don't write nλ or (n + ½)λ there.
Writing a list of facts in a 6-mark answer. Link each step with "so" or "because".

BEYOND THE SPEC, BUT EXAMINED

- **Noise-cancelling headphones**: the cancelling wave must arrive in antiphase, giving destructive interference (WPH12 Oct 2020).
- **Beats**: two notes of 880 Hz and 882 Hz are not coherent, so their phase difference drifts; the sound is loud when they are in phase and quiet in antiphase (WPH12 Oct 2022). Answer in terms of phase, not path difference.
- **Thin film**: light reflected from the top and bottom of a coating of thickness d has path difference 2d. Use the wavelength **inside** the coating, λ_air ÷ n (WPH12 Oct 2023).

SPEC 40

Phase difference and path difference

WHAT THIS SPEC POINT ASKS

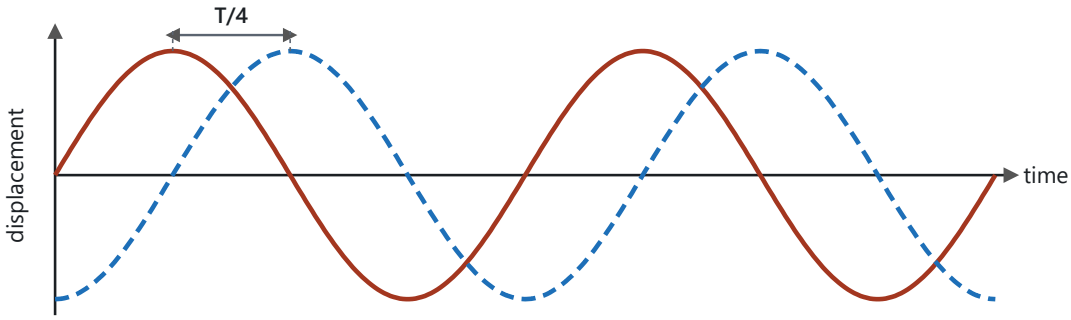
Convert between the phase difference and the path difference of two waves.

40 The relationship

One whole wavelength of path difference = one whole cycle = 360° = 2π rad. So:

$$\text{phase difference} = (\text{path difference} / \lambda) \times 360° = (\text{path difference} / \lambda) \times 2\pi \text{ rad}$$

The same idea works for two points on one progressive wave a distance Δx apart (use Δx as the "path difference"), and for time: phase difference = (Δt / T) × 360°.



Solid A, dashed B. B lags A by a quarter cycle: phase difference = 90° = π/2 rad

Two oscillations a quarter of a period (or quarter of a wavelength) apart have a phase difference of 90°, π/2 rad.

Path difference	Phase difference	Interference
0, λ, 2λ	0 (or 360°, 720°)	constructive
λ/4	90° = π/2 rad	neither
λ/2, 3λ/2	180° = π rad	destructive
3λ/4	270° = 3π/2 rad	neither

WORKED EXAMPLE: TWO LOUDSPEAKERS

Two loudspeakers driven by the same 500 Hz signal. Speed of sound 340 m s^{-1}. A point P is 4.10 m from one speaker and 3.08 m from the other.
λ = v/f = 340 ÷ 500 = 0.680 m.
Path difference = 4.10 − 3.08 = 1.02 m = 1.02 ÷ 0.680 = 1.5 λ.
Phase difference = 1.5 × 360° = 540°, which is the same as **180° (π rad)**: antiphase.
So destructive interference: P is a **quiet** point (silent only if the two amplitudes are equal).
Two points on one of these waves 0.17 m apart: 0.17 ÷ 0.680 = 0.25 λ, so phase difference = **90° (π/2 rad)**.

MARK-SCHEME LANGUAGE

- "Phase difference = 2π × (path difference)/λ" or the degree version, with the answer as an **angle**.
- Whole number of wavelengths → in phase; odd number of half wavelengths → antiphase.

Often asked as: one-mark MCQs converting λ fractions to angles (3λ/8 = 135°; λ/8 = π/4; 7λ/4 = 3.5π rad, the same as 1.5π rad).

COMMON MISTAKE

Forgetting to subtract whole cycles: 540° and 180° describe the same phase relationship.
Mixing degrees and radians in one line. Pick one.

BEYOND THE SPEC, BUT EXAMINED

One mark scheme (WPH12 Oct 2019, MCQ) wrote that 7λ/4 is "3.5 radians". It is **3.5π rad** (equivalent to 1.5π rad). The answer letter was right; the working was not. Don't copy it.

SPEC 41 Stationary (standing) waves

WHAT THIS SPEC POINT ASKS

Know what a stationary wave is and how one forms, and identify its nodes and antinodes.

41 How one forms

1. A wave travels along a string (or through air in a pipe) and is **reflected** at the end (the fixed end, the bridge, the closed end of a pipe, a metal plate…).
2. The incident and reflected waves have the **same frequency** (and similar amplitude) and travel in **opposite directions**.
3. They **superpose**. At some points they always cancel (**nodes**); at others they always reinforce (**antinodes**).
4. A steady pattern only appears at certain frequencies (the resonant frequencies), when a whole number of half-wavelengths fits the length.

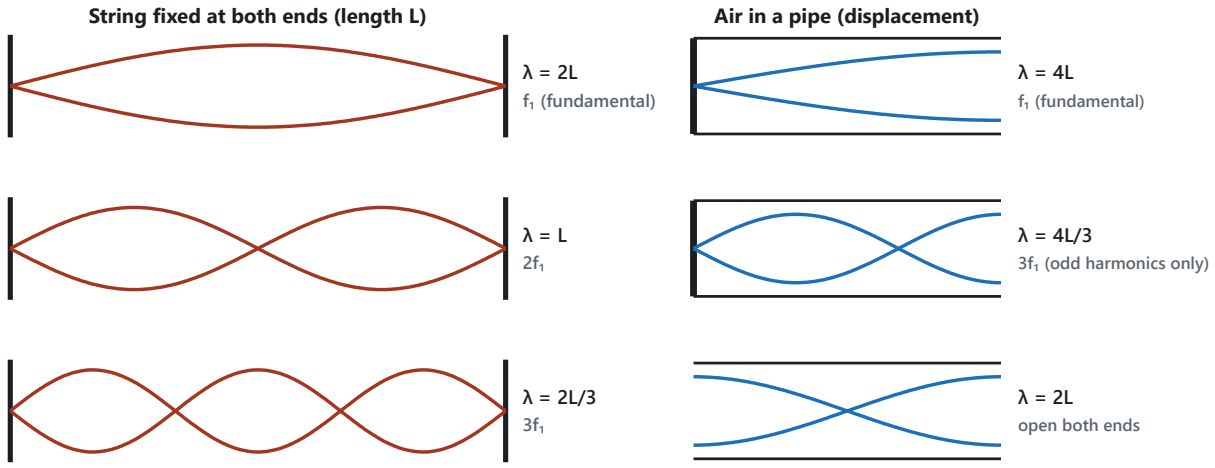
41 Nodes and antinodes

- **Node:** a point of **zero amplitude**. The two waves always arrive in antiphase: **destructive** interference.
- **Antinode:** a point of **maximum amplitude**. The two waves always arrive in phase: **constructive** interference. The displacement at an antinode still varies, from +A through zero to −A.
- Node to next node = **λ/2**. Node to next antinode = **λ/4**.

41 Stationary v progressive

	Progressive wave	**Stationary wave**
Energy	transferred along the wave	**not transferred** (stored in the oscillation)
Amplitude	the same at every point (no losses)	varies with position: zero at nodes, max at antinodes
Phase	changes steadily along the wave over one λ	all points between adjacent nodes in phase; adjacent loops in antiphase
Wavelength	distance between adjacent points in phase	twice the node-to-node distance

41 Harmonics on strings and in pipes



Closed end = node. Open end = antinode. Fixed string end = node.

A string fixed at both ends has a node at each end. A pipe closed at one end has a node at the closed end and an antinode at the open end. A pipe open at both ends has antinodes at both ends.

System	Fundamental (first harmonic)	Higher harmonics
String fixed at both ends, length L	$\lambda = 2L$	$\lambda = 2L/n$, $f = n f_1$ ($n = 1, 2, 3, \ldots$)
Pipe closed at one end	$\lambda = 4L$	odd only: $3f_1$, $5f_1$, …
Pipe open at both ends	$\lambda = 2L$	all: $2f_1$, $3f_1$, …

41 Measuring λ with microwaves

Point a microwave transmitter at a metal plate. Move a detector along the line between them: the reading rises and falls as it passes antinodes and nodes. Measure across several nodes: node-to-node distance = λ/2.

WORKED EXAMPLE: WAVELENGTH FROM NODES

A microwave detector finds 6 nodes spread over 7.0 cm (that is 5 node-to-node gaps).
λ/2 = 7.0 ÷ 5 = 1.4 cm, so λ = 2.8 cm = 0.028 m.
$f = c/\lambda = 3.00 \times 10^8 \div 0.028 =$ **1.1×10^{10} Hz** (2 s.f.).
A string 1.70 m long showing 4 loops: 4 × λ/2 = 1.70 m, so λ = **0.85 m** (WPH12 Jan 2020 style).

MARK-SCHEME LANGUAGE

- Formation: "wave **reflected** at [the named reflector]", "two waves travelling in **opposite directions**" (or "reflected back on itself"), "same frequency", "**superposition / interference**".
- Node: "zero / minimum amplitude", "destructive interference". Antinode: "maximum amplitude", "constructive interference".
- "Node to node = λ/2."

Often asked as: explain how the stationary wave forms in a context (string, tube above water, bridge cable, metal plate), distinguish nodes and antinodes, λ from a pattern, tube resonance frequencies.

COMMON MISTAKE

Quoting a textbook definition without the context. The first mark often needs the actual reflector, e.g. "reflected by the water surface"; very few candidates said this (WPH12 Oct 2021 examiner report).
"Two opposite waves". Say "travelling in opposite directions".
Giving λ as twice the string length when there are several loops (WPH12 Jan 2020 examiner report). Count the loops first.

BEYOND THE SPEC, BUT EXAMINED

- **Closed tube resonance**: a tube above water, or closed at one end, resonates first when its length is λ/4, so λ = 4L. E.g. 440 Hz × (4 × 0.193 m) = 340 m s^{-1} (WPH12 Oct 2021).
- **End correction**: the antinode sits slightly beyond the open end of a tube, so the real λ is a bit longer than 4L and a speed worked out from 4L comes out too low (WPH12 Jun 2025).
- To lower the pitch of a recorder-type instrument (open pipe, λ = 2L), make the air column longer (WPH12 Jan 2021).

SPEC 42

Speed of a transverse wave on a string

WHAT THIS SPEC POINT ASKS

Use $v = \sqrt{(T/\mu)}$ for the speed of a transverse wave on a stretched string.

42 The equation

$$v = \sqrt{(T / \mu)}$$

- v = wave speed on the string (m s^{-1}); T = **tension** (N); μ = **mass per unit length** (kg m^{-1}) = mass of the string ÷ its length. It is on the formulae sheet.
- Tighter string → faster wave. Heavier string → slower wave.
- If the tension comes from a hanging mass m: **T = mg**, with g = 9.81 N kg^{-1}.

42 Linking it to frequency

For the fundamental of a string fixed at both ends, λ = 2L (L = **vibrating** length, between the two fixed points). Using v = fλ:

$$f = (1 / 2L)\sqrt{(T / \mu)}$$

Change	Effect on f_1
double the length L	f halves ($f \propto 1/L$)
quadruple the tension T	f doubles ($f \propto \sqrt{T}$)
quadruple μ (heavier string)	f halves ($f \propto 1/\sqrt{\mu}$)

WORKED EXAMPLE: FREQUENCY OF A LOADED STRING

A string with μ = 1.20 × 10^{-3} kg m^{-1} passes over a pulley and holds a 0.500 kg mass. The vibrating length is 0.800 m.
T = mg = 0.500 × 9.81 = 4.905 N.
$v = \sqrt{(T/\mu)} = \sqrt{(4.905 \div 1.20 \times 10^{-3})} = \sqrt{4088} = 63.9$ m s^{-1}.
Fundamental: λ = 2L = 1.60 m, so f = v/λ = 63.9 ÷ 1.60 = **40.0 Hz** (3 s.f.).

WORKED EXAMPLE: TUNING BY TENSION

A string sounds 432 Hz. What percentage increase in tension raises it to 440 Hz?
$f \propto \sqrt{T}$, so $T \propto f^2$. New T ÷ old T = $(440 \div 432)^2$ = 1.037. The tension must rise by about **3.7 %**.

MARK-SCHEME LANGUAGE

- "Use of T = mg", "use of $v = \sqrt{(T/\mu)}$", "recognises $\lambda = 2L$", then "use of $v = f\lambda$".
- Ratio questions: "$f \propto \sqrt{T}$" so "$T \propto f^2$".

Often asked as: speed and frequency of a loaded string, choose which string gives a required note, small tension changes, MCQs on what changes v.

COMMON MISTAKE

Multiplying μ by the length. μ is already "per metre" (WPH12 Jan 2020 examiner report). If you're given a total mass, μ = m ÷ L.
Using g = 10 N kg^{-1} costs a mark. Use 9.81.
Using the full string length when only part of it vibrates (bridge to bridge).
No unit on the final answer.

BEYOND THE SPEC, BUT EXAMINED

Bridge cables: μ = cable mass ÷ cable length (e.g. 2.10×10^3 kg ÷ 87.0 m = 24.1 kg m^{-1}), and the tension is the load shared between the cables. You then show which frequencies the cable can support (WPH12 Oct 2024).

SPEC 43

Core practical 5: frequency of a vibrating string

WHAT THIS SPEC POINT ASKS

Investigate how the length, tension and mass per unit length of a string affect its frequency of vibration.

CORE PRACTICAL 5: METHOD IN BRIEF

1. Fix one end of a string to a **vibration generator** driven by a **signal generator**. Run the string over a bridge and a pulley, with a mass hanger on the end (tension T = mg).
2. Set a vibrating length L between the vibration generator (or a bridge) and the pulley bridge. Measure it with a metre rule.
3. Slowly increase the frequency from low until the string shows **one loop of maximum amplitude** (the fundamental). Record f from the signal generator.
4. **Length:** move the bridge to change L; keep T and μ fixed.
5. **Tension:** add masses; keep L and μ fixed.
6. **Mass per unit length:** use different strings; find each μ by weighing a measured long length (mass ÷ length); keep L and T fixed.
7. Repeat each reading (approach the resonance from above and below) and average.

43 Turning the results into straight lines

From $f = (1/2L)\sqrt{(T/\mu)}$:

Investigation	Plot	Straight line through origin, gradient
Length	f against 1/L	½√(T/μ)
Tension	f^2 against T	$1/(4L^2\mu)$
Mass per unit length	f against $1/\sqrt{\mu}$	$(1/2L)\sqrt{T}$

43 Errors and safety

- Judging the exact point of maximum amplitude is hard. Approach it from both sides and average; or measure across several loops at a higher harmonic.
- **Parallax** when reading L: hold the ruler close to the string and read at eye level.
- The generator dial may be inaccurate: check f with an oscilloscope (measure T, f = 1/T).
- Safety: masses can fall. Put a cushion or box underneath and keep feet clear. A snapping string can whip: wear eye protection.

WORKED EXAMPLE: M FROM AN F^2 AGAINST T GRAPH

With L = 0.600 m, a graph of f^2 against T has gradient 1.80×10^3 Hz^2 N^{-1}.
Gradient = $1/(4L^2\mu)$, so $\mu = 1 \div (4 \times 0.600^2 \times 1.80 \times 10^3) = 1 \div 2592 =$ **3.86×10^{-4} kg m^{-1}** (3 s.f.).

MARK-SCHEME LANGUAGE

- "Show that $f^2 \propto W$" (5 marks): equate $v = f\lambda$ and $v = \sqrt{(T/\mu)}$; rearrange to $f^2 = T/(\mu\lambda^2)$ or $T/(4L^2\mu)$; T = W; "μ and λ (or L) are constant"; so $f^2 \propto W$, "y = mx with no c / y-intercept is 0" (WPH12 Oct 2020).
- Measuring f: "read from the signal generator" or "use an oscilloscope: measure T, f = 1/T".
- Uncertainty: "difficult to judge the exact position of the node"; "ruler not close to the string, so parallax" (WPH12 Jan 2022).

Often asked as: show a graph is a straight line through the origin, find μ or a mass from a gradient, how to measure f, sources of uncertainty, MCQ on which variable is not a control.

BEYOND THE SPEC, BUT EXAMINED

Turning an unfamiliar equation with two squared quantities into y = mx + c and reading μ from the gradient, e.g. f^2 against T with gradient 532 Hz^2 N^{-1} giving $\mu = 6.5 \times 10^{-4}$ kg m^{-1} (WPH12 Jan 2025).

SPEC 44 Intensity of radiation, I = P/A

WHAT THIS SPEC POINT ASKS

Use I = P/A for the intensity of radiation.

44 Definition

Intensity is the power transferred per unit area, through an area **perpendicular** to the direction the wave travels.

$$I = P / A$$

I = intensity (W m^{-2}), P = power (W), A = area (m^2). It is on the formulae sheet.

44 The inverse square law

A point source sends its power out equally in all directions. At distance r the power is spread over the surface of a sphere, **$A = 4\pi r^2$**:

$$I = P / 4\pi r^2$$

- Double the distance → intensity drops to **a quarter**. Triple it → a ninth.
- This assumes no energy is absorbed on the way.
- Intensity also rises with the square of the amplitude (twice the amplitude, four times the intensity).

WORKED EXAMPLE: POINT SOURCE AND THE SUN

(a) A small speaker emits 0.50 W of sound equally in all directions. At 3.0 m:
$I = P/4\pi r^2 = 0.50 \div (4\pi \times 3.0^2) = 0.50 \div 113 =$ **4.4×10^{-3} W m^{-2}** (2 s.f.). At 6.0 m it is a quarter of this, 1.1×10^{-3} W m^{-2}.
(b) Sunlight at the Earth (1.50×10^{11} m from the Sun) has intensity 1.36×10^3 W m^{-2}. Power of the Sun:
$P = I \times 4\pi r^2 = 1.36 \times 10^3 \times 4\pi \times (1.50 \times 10^{11})^2 =$ **3.85×10^{26} W** (3 s.f.).

WORKED EXAMPLE: TILTED SOLAR PANEL

Sunlight of intensity 1.00 kW m^{-2} falls on a 1.60 m^2 panel. The light meets the panel at 30.0° to its normal. The panel is 18.0 % efficient.
Effective area facing the Sun = $A \cos\theta = 1.60 \times \cos 30.0° = 1.386$ m^2.
Power arriving = $1.00 \times 10^3 \times 1.386 = 1386$ W.
Useful output = $0.180 \times 1386 =$ **249 W** (3 s.f.).

MARK-SCHEME LANGUAGE

- "Use of I = P/A", "use of $A = 4\pi r^2$" (a dimensionally correct area: not πr^2 for a sphere), then compare with the quoted value and conclude.
- Intensity definition: "power per unit area" or "energy per second per metre squared".
- MCQ: power constant, so I × A is constant; doubling distance quarters I.

Often asked as: on almost every paper: stars and the Sun, routers, LEDs, solar panels with efficiency, number of photons per second from an intensity (58).

COMMON MISTAKE

Using πr^2 instead of $4\pi r^2$ for a point source.
Not converting cm^2 to m^2 (1 cm^2 = 10^{-4} m^2). This turned 0.48 W into 48 W for many (WPH12 Jan 2025 examiner report).
Comparing different time spans (a year's energy against one hour's) or giving a power when an energy is asked for. Use E = Pt and convert hours and years to seconds.

BEYOND THE SPEC, BUT EXAMINED

The **cos θ** factor for a tilted panel and **efficiency = useful power out ÷ total power in** appear regularly, e.g. 1.1 kW m^{-2} × 2.2 m^2 × cos 57° = 1.32 kW, of which 20 % is used (WPH12 Jun 2026).

SPEC 45

Refraction and refractive index

WHAT THIS SPEC POINT ASKS

Know that $n_1 \sin \theta_1 = n_2 \sin \theta_2$ at the boundary between two media, where the refractive index $n = c/v$.

45 What refraction is

- **Refraction** is the change in **direction** of a wave when it crosses a boundary between two media, caused by a change in its **speed**.
- Light slows down entering a more optically dense medium (higher n) and bends **towards the normal**. It speeds up leaving it and bends **away from the normal**.
- A ray along the normal (angle of incidence 0°) doesn't change direction, but its speed still changes.
- **Why it bends:** a wavefront meeting the boundary at an angle has one side enter the new medium first. That side slows down first while the other side is still moving fast, so the wavefront swings round.
- Frequency stays the same; wavelength changes with the speed ($n_1\lambda_1 = n_2\lambda_2$).

45 The equations

$$n = c / v$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

- n = absolute refractive index (no unit), c = 3.00×10^8 m s^{-1} (speed of light in a vacuum), v = speed of light in the material.
- n is always ≥ 1. Air ≈ 1.00, water 1.33, glass about 1.5.
- θ_1, θ_2 are the angles to the **normal** in medium 1 (where the light comes from) and medium 2.
- Both are on the formulae sheet.

WORKED EXAMPLE: SNELL'S LAW THREE WAYS

(a) Light in air meets water (n = 1.33) at 40.0° to the normal.
$1.00 \times \sin 40.0° = 1.33 \times \sin \theta_2 \rightarrow \sin \theta_2 = 0.6428 \div 1.33 = 0.4833 \rightarrow \theta_2 =$ **28.9°**.
(b) Speed of light in water: $v = c/n = 3.00 \times 10^8 \div 1.33 =$ **2.26×10^8 m s^{-1}** (3 s.f.).
(c) Light passes from glass (n = 1.50) into water at 30.0° to the normal.
$1.50 \times \sin 30.0° = 1.33 \times \sin \theta_2 \rightarrow \sin \theta_2 = 0.750 \div 1.33 = 0.5639 \rightarrow \theta_2 =$ **34.3°**. It bends away from the normal because water has the lower n.

MARK-SCHEME LANGUAGE

- "Use of $n_1 \sin \theta_1 = n_2 \sin \theta_2$ with both values seen" (e.g. 1.33 and 1.52 both visible). n = sin i / sin r is allowed from air.
- Explain refraction: "the wave slows down as it enters the glass"; "part of the wavefront reaches the glass first, so the direction changes". "Density/refractive index of glass greater" is accepted as the reason.
- Define: "change of direction of a ray due to a change in speed/medium". "Bending" alone is not accepted (WPH12 Jan 2021).

Often asked as: on nearly every paper: angles measured from a diagram, n from angles then the speed, identify a gemstone from a table, explain refraction with wavefronts.

COMMON MISTAKE

Measuring angles from the **surface** instead of the normal (or reading the wrong protractor scale, 35° for 55°) (WPH12 Jun 2022 examiner report).
Putting **speeds** into the $n_1 \sin \theta_1 = n_2 \sin \theta_2$ slots (scored 0, WPH12 Oct 2021 examiner report). Convert to n first with $n = c/v$.
Rounding to 2 s.f. in the middle of a dispersion question: the small difference between colours vanishes. Keep 4 s.f. until the end.

BEYOND THE SPEC, BUT EXAMINED

- **Dispersion**: n is slightly larger for violet than red, so violet bends more and has a smaller critical angle (WPH12 Jun 2021).
- **Brewster angle**: when the reflected and refracted rays are at 90° to each other, Snell's law gives $\tan \theta_B = n_2/n_1$ (3-mark derivation, WPH12 Oct 2019). The reflected light is then plane polarised.
- A wavefront parallel to the boundary enters all at once, so there's no change in direction (WPH12 Oct 2023).

SPEC 46

Critical angle

WHAT THIS SPEC POINT ASKS

Calculate the critical angle using sin C = 1/n.

46 Definition and derivation

- The **critical angle C** is the angle of incidence, in the **more optically dense** medium, for which the angle of refraction is **90°** (the refracted ray runs along the boundary).
- From Snell's law with $\theta_1 = C$ and $\theta_2 = 90°$ ($\sin 90° = 1$): $n_1 \sin C = n_2$.

$$\sin C = n_2 / n_1 \text{ and, into air } (n_2 = 1.00): \sin C = 1 / n$$

- It only exists going from higher n to lower n. The bigger the difference in n, the smaller C.

WORKED EXAMPLE: CRITICAL ANGLES

(a) Glass–air, n = 1.50: $\sin C = 1 \div 1.50 = 0.6667 \rightarrow C =$ **41.8°**.
(b) Glass (1.50)–water (1.33): $\sin C = 1.33 \div 1.50 = 0.8867 \rightarrow C =$ **62.5°**. Water around the glass makes TIR harder.
(c) Optical fibre, core 1.48, cladding 1.43: $\sin C = 1.43 \div 1.48 = 0.9662 \rightarrow C =$ **75.1°**.

MARK-SCHEME LANGUAGE

- "Use of sin C = 1/n" or "$n_1 \sin C = n_2 \sin 90°$" with both values seen.
- If n isn't given, find it first from speeds, n = c/v (e.g. n = 1.53, C = 40.8°, WPH12 Jun 2022).

Often asked as: calculate C for a pair of materials, use C in a geometry problem (prism or semicircular block), MCQs on the conditions.

COMMON MISTAKE

Using the refractive index of a different boundary from an earlier part of the question (WPH12 Jun 2022 examiner report).
Working out C correctly and then drawing the ray refracting out anyway (WPH12 Oct 2021 examiner report).

SPEC 47 Predicting total internal reflection

WHAT THIS SPEC POINT ASKS

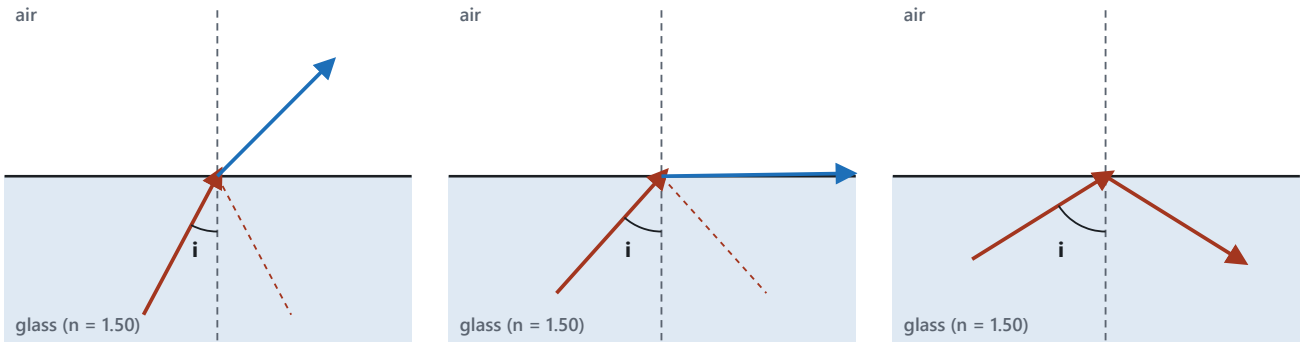
Predict whether total internal reflection will happen at a boundary.

47 The two conditions

Total internal reflection (TIR) happens when **both** are true:

1. The light is going from a medium of **higher** refractive index to one of **lower** refractive index (more to less optically dense).
2. The angle of incidence is **greater than the critical angle**.

Then **all** the light is reflected back into the first medium. Below C, the light mostly refracts out but a weak partial reflection always happens too.



i < C: refracts out (+ weak reflection) | i = C: refracted ray along the surface | i > C: total internal reflection

Glass to air (C = 41.8°). Left: refraction plus a weak reflection. Middle: at C the refracted ray runs along the surface. Right: above C, all the light is reflected.

47 Two ways to predict

- **Compare i with C.** Work out C for the boundary; if i > C, TIR.
- **Try Snell's law.** If $n_1 \sin\theta_1 \div n_2$ comes out bigger than 1, there's no possible refraction angle (sin can't exceed 1), so TIR. Mark schemes accept this.

47 Optical fibres

- A glass **core** surrounded by **cladding** of slightly lower n. Light hits the core–cladding boundary above C, so it is totally internally reflected along the fibre.
- The cladding protects the core from scratches and stops light leaking into neighbouring fibres. Because it has n > 1, it **raises** the critical angle compared with air.
- Rays at different angles travel different path lengths, so they arrive at different times and a short pulse **spreads out** (and some rays refract out, so the signal weakens).

WORKED EXAMPLE: WILL TIR HAPPEN?

(a) Light in a 45° glass prism (n = 1.50) hits the long face at 45°. C = 41.8°. 45° > 41.8°, so **TIR**.
(b) Light in water (n = 1.33, C = 48.8°) meets the water surface at 45°. 45° < 48.8°, so it refracts out: $\sin \theta_2 = 1.33 \times \sin 45° = 0.9405$, $\theta_2 =$ **70.1°**.
(c) The same water, ray at 55°: $1.33 \times \sin 55° = 1.09$. That is more than 1, so no refracted ray: **TIR**.

MARK-SCHEME LANGUAGE

- Conditions: "travelling from a more optically dense to a less optically dense medium" and "angle of incidence greater than (or equal to) the critical angle".
- Say "**total internal reflection**". "Reflection" alone, or "bending" for refraction, doesn't score (WPH12 Oct 2024 examiner report).
- Fibres: "different path lengths, so different times, so the pulse is longer"; "lower cladding index gives a lower critical angle, so more light is totally internally reflected".

Often asked as: compare i with C and draw the ray, prisms (some deliberately not 45°), 6-mark ray-picture reasoning, optical fibres.

COMMON MISTAKE

Assuming a prism is 45° without checking the angles. Work out the angle of incidence from the geometry (WPH12 Oct 2021 examiner report).
Forgetting the first condition: light going from air into glass can never be totally internally reflected.

BEYOND THE SPEC, BUT EXAMINED

Fingerprint scanners: where a skin ridge touches the glass, light is partly transmitted into the skin (dark); at air-filled valleys it is totally internally reflected (bright) (WPH12 Oct 2023).

SPEC 48

Measuring the refractive index of a solid

WHAT THIS SPEC POINT ASKS

Understand how to measure the refractive index of a solid material.

MEASURING N WITH A GLASS BLOCK: METHOD IN BRIEF

1. Put a rectangular glass (or Perspex) block on paper and draw round it.
2. Shine a **narrow** ray from a ray box into one face at an angle. Mark the ray going in and coming out with small crosses far apart.
3. Remove the block. Join the crosses and the entry and exit points to draw the ray inside the block.
4. Draw the **normal** at the entry point with a protractor. Measure the angle of incidence i and angle of refraction r **from the normal**.
5. Repeat for 5–6 angles of incidence (e.g. 20° to 70°).
6. Plot **sin i (y) against sin r (x)**. The gradient is **n** (from air, n = sin i / sin r).

Semicircular block (critical angle): aim the ray at the centre of the flat face through the curved side, so it doesn't bend on entering. Increase the angle until the refracted ray just grazes the flat face. That angle is C, and n = 1/sin C.

48 Errors and improvements

- A wide ray makes the angles uncertain: use a narrow slit and mark the **centre** of the ray.
- A protractor reads to ±1°. Larger angles give a smaller percentage uncertainty.
- A graph of many readings is better than one pair: it averages random errors and shows anomalies.
- **Liquids**: put the liquid in a thin-walled transparent container (the walls make little difference) and do the same.

WORKED EXAMPLE: N FROM ONE PAIR OF ANGLES

From air into a block: i = 50.0°, r = 30.5°.
n = sin i / sin r = 0.7660 ÷ 0.5075 = **1.51** (3 s.f.).
If a question gives distances instead of angles (e.g. a ray crossing a block of known width), find tan r = opposite/adjacent first, then r, then use Snell's law.

MARK-SCHEME LANGUAGE

- "Use of trigonometry to correctly determine either i or r", then Snell's law with those angles (WPH12 Jan 2021). Show the trig: an angle read with a protractor doesn't earn that mark.
- Graph: "gradient of sin i against sin r = n".

Often asked as: not as a method question so far, but through ray diagrams and measured results: measure the angles on a printed diagram, use distances to find angles, semicircular blocks.

COMMON MISTAKE

Diagrams in questions are often drawn to scale and mark schemes accept a range (e.g. 54°–56°). Measure carefully from the normal; don't assume a "nice" angle.

SPEC 49 Plane polarisation

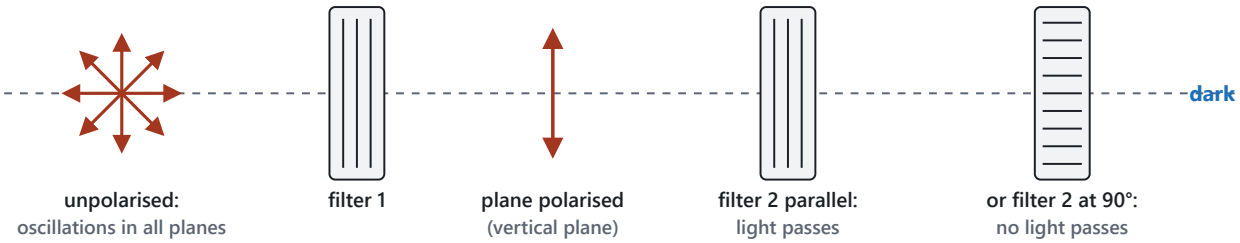
WHAT THIS SPEC POINT ASKS

Understand what plane polarisation means.

49 Unpolarised and plane polarised

- In **unpolarised** light (e.g. from the Sun or a lamp) the oscillations are in **many planes**, all containing the direction of travel.
- In **plane polarised** light the oscillations are in **one plane only**, which contains the direction of travel.
- Only **transverse** waves can be polarised. Sound can't be: its oscillations are already along the direction of travel.

49 Polarising filters



Filter 1 lets through only the oscillations in one plane. A second filter in the same orientation passes it; at 90° (crossed) it blocks it all.

- A polarising filter transmits only oscillations in one plane. Unpolarised light that passes through comes out plane polarised, at about half the intensity.
- **Testing whether light is polarised:** look through one filter and rotate it.
  - Intensity changes from maximum to minimum **every 90°** → the light is plane polarised. Maximum when the filter is lined up with the plane of oscillation, minimum (zero) when it is at 90°.
  - Intensity doesn't change → the light is unpolarised.

49 Where polarisation turns up

- **Reflection**: light reflected from water, glass or a road is **partially polarised** (mostly horizontally). Polarising sunglasses block that plane and cut the glare.
- **Screens** (LCDs, phones) emit polarised light. Turn the phone through 90° while wearing polarising sunglasses and the screen goes dark.
- **Stress analysis**: put a clear plastic model between **crossed** filters. Stress rotates the plane of polarisation, so light gets through the second filter; the brighter areas (with closely spaced coloured fringes) show the greatest stress.
- **Radio and TV aerials** must line up with the transmitter's plane of polarisation.

MARK-SCHEME LANGUAGE

Two accepted versions. Use one consistently; don't mix them:
- "Unpolarised: oscillations in **all/many planes**. Polarised: oscillations in **one plane** which **includes** the direction of wave travel."
- Or: "Unpolarised: oscillations in all directions. Polarised: oscillations in **one direction**, **perpendicular** to the direction of wave travel."
- "Direction of energy transfer/propagation" is accepted for direction of travel. The last mark depends on the first ones.
- Rotation: "maximum to minimum every 90°"; "brightness" or "amount of light" accepted for intensity.

Often asked as: define plane polarised (2–3 marks) on many papers, test for polarisation, reflected light, sunglasses and phone screens, stress analysis, three filters.

COMMON MISTAKE

Reciting the definition when the question is about a context. Describe what the filter does in **that** situation (WPH12 Jun 2024 examiner report).
Answering a one-filter question with the two-filter answer (WPH12 Jun 2019 examiner report).
"In one plane" without saying the plane includes the direction of travel.

BEYOND THE SPEC, BUT EXAMINED

- **Three filters**: filters 1 and 2 aligned, filter 3 at 90° to filter 2: no light. Rotate filter 2 to an in-between angle and some light now passes filter 3, so the intensity **increases** (WPH12 Oct 2025).
- **Intensity against angle** sketch for a rotating filter in polarised light: one maximum at 0°, zero at 90°, a smooth curve between (a $\cos^2$ shape; the equation isn't needed) (WPH12 Jun 2021).

SPEC 50

Diffraction and Huygens' construction

WHAT THIS SPEC POINT ASKS

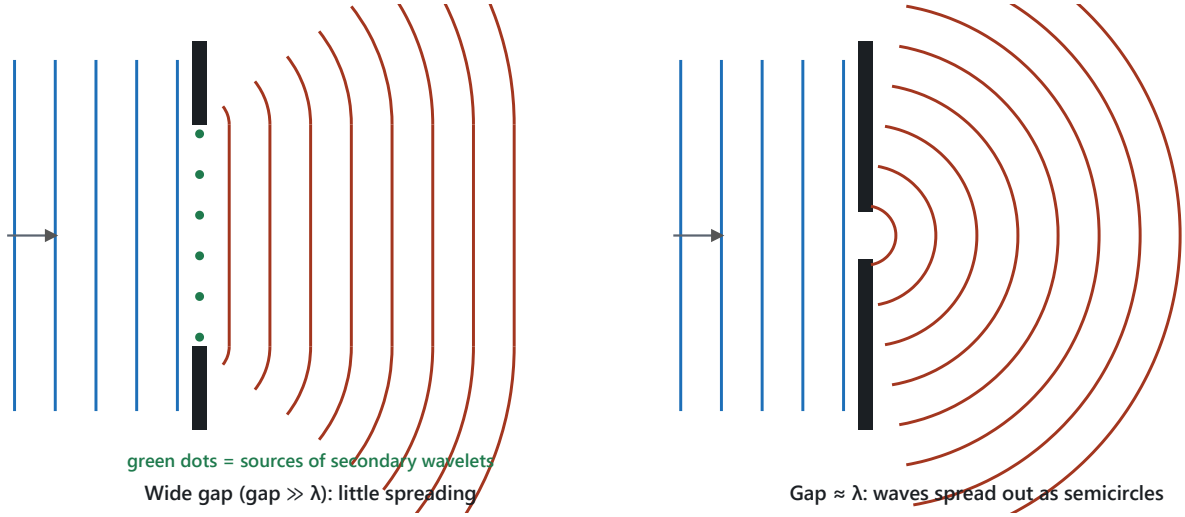
Know what diffraction is, and use Huygens' construction to explain what happens when a wave meets a slit or an obstacle.

50 Diffraction

- **Diffraction** is the **spreading out** of waves when they pass through a gap or around an obstacle.
- The **wavelength, frequency and speed don't change**. The energy is spread over a wider area, so the amplitude falls.
- The effect is **greatest when the gap is about the same size as the wavelength**. A gap much wider than λ gives little spreading.
- Sound (λ ≈ 1 m) diffracts noticeably round doorways; visible light (λ ≈ 5 × 10^{-7} m) needs a very narrow slit to show it.
- Diffraction is a **wave** property. Seeing it is evidence that something behaves as a wave (53).

50 Huygens' construction

1. Treat **every point on a wavefront** as a source of **secondary wavelets** that spread out at the wave speed.
2. A short time later, the new wavefront is the line that touches all the wavelets (their **envelope**).
3. In the middle of a wide gap the wavelets add up to a straight wavefront. At the **edges** there are no neighbouring wavelets beyond the gap, so the wavefront curves round: that is the spreading.
4. If the gap is about λ wide, it acts almost like a single source, so the waves spread out as semicircles.



Same wavelength before and after the gap. A wide gap only curves the edges; a gap about one wavelength wide spreads the waves out into semicircles.

50 Single-slit pattern for light

Monochromatic light through one narrow slit gives a **wide, bright central maximum** with much dimmer, narrower maxima either side. Narrow the slit, or use longer λ, and the pattern spreads further.

MARK-SCHEME LANGUAGE

- "Waves spread out as they pass through a gap or around an obstacle" ("slit" accepted).
- Huygens: "each point on a wavefront acts as a source of secondary wavelets"; "the wavelets superpose/interfere to form the new wavefront".
- Patterns: "diffraction **and** interference take place"; maxima = constructive, minima = destructive.
- Resolution: "higher frequency, shorter wavelength, so **less diffraction**, so more detail" (see 56).

Often asked as: define diffraction, describe Huygens' construction, sketch a single-slit intensity graph, explain bright and dark patches, why short wavelengths give better resolution.

COMMON MISTAKE

Saying the wavelength changes after the gap. It doesn't.
Saying diffraction "needs" a gap equal to λ. It happens with any gap; it is just most noticeable when the gap ≈ λ.

BEYOND THE SPEC, BUT EXAMINED

- **Single-slit sketch**: central maximum halfway between the edges, **twice as wide** as the side maxima and much taller (WPH12 Jan 2020).
- **Two gaps, 6 marks**: waves from each gap diffract and overlap; at the centre there's no path difference so constructive; elsewhere a whole number of λ of path difference gives constructive and an odd number of half wavelengths gives destructive, so with mixed wavelengths different colours peak in different places (WPH12 Jan 2024).

SPEC 51 Diffraction grating, nλ = d sin θ

WHAT THIS SPEC POINT ASKS

Use nλ = d sin θ for a diffraction grating.

51 What a grating does

- A **diffraction grating** is a plate with a very large number of **equally spaced, parallel slits**.
- Light diffracts through every slit and the waves interfere. Because there are so many slits, the maxima are **very sharp and bright**, at precise angles, so angles can be measured accurately.

51 Where the equation comes from

Take two neighbouring slits a distance d apart. At angle θ, the wave from one has to travel an extra **d sin θ**. Bright maxima appear where this path difference is a whole number of wavelengths, so the waves from **all** the slits arrive in phase:

$$n\lambda = d \sin\theta$$

- n = order of the maximum (0, 1, 2, …; the straight-through spot is n = 0); λ = wavelength (m); d = slit spacing (m); θ = angle from the **straight-through (zero-order) direction**. It is on the formulae sheet.

- **d = 1 ÷ (number of lines per metre).** 300 lines per mm = 3.00×10^5 lines per metre, so d = 3.33×10^{-6} m.
- **Highest order:** sin θ can't exceed 1, so n ≤ d/λ. Round **down** to a whole number.
- **White light:** a white central maximum, then a spectrum in each order with **violet nearest the centre** (shortest λ, smallest θ) and red furthest out. Higher orders can overlap.

WORKED EXAMPLE: ANGLES AND THE NUMBER OF MAXIMA

A grating has 300 lines per mm and is lit by a 633 nm laser.
d = 1 ÷ (300×10^3) = 3.33×10^{-6} m.
First order: sin θ = λ/d = $6.33 \times 10^{-7} \div 3.33 \times 10^{-6} = 0.190$ → θ = **10.9°**.
Second order: sin θ = 2 × 0.190 = 0.380 → θ = **22.3°**.
Highest order: n ≤ d/λ = $3.33 \times 10^{-6} \div 6.33 \times 10^{-7} = 5.27$, so n = 5. Maxima seen: 5 each side + the central one = **11**.

MARK-SCHEME LANGUAGE

- "Use of nλ = d sin θ". Single- or double-slit equations get no credit (WPH12 Jun 2019).
- Why maxima form: "path difference between waves from **adjacent slits** is a whole number of wavelengths, so they are in phase, so constructive interference, giving **maxima**".
- "Allow a reverse calculation" to check a label (e.g. 300 lines per mm), but the conclusion must compare with the label.

Often asked as: very common: find d or lines per mm, find λ, highest order, what happens if d or λ changes, why bright dots appear, white-light spectra.

COMMON MISTAKE

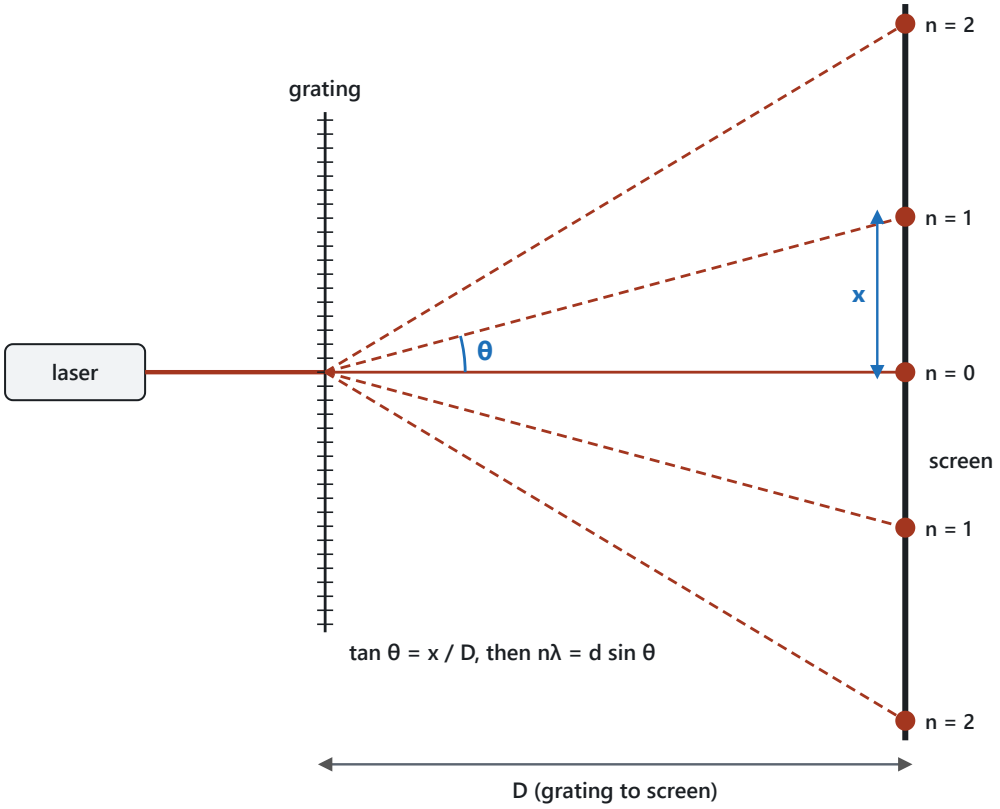
Using the angle between the two first orders as θ. **Halve it** (WPH12 Oct 2024 examiner report).
Mixing up n and d, or slipping a power of ten when converting lines per mm.
Taking the reciprocal of a slit spacing that was already given (WPH12 Jun 2024 examiner report).
Using tan θ = x/D and then putting tan θ into the grating equation. Find θ, then take sin θ.
Generic "constructive and destructive interference" without explaining why the maxima are at those angles.

SPEC 52

Core practical 6: wavelength of light with a diffraction grating

WHAT THIS SPEC POINT ASKS

Find the wavelength of light from a laser (or another source) using a diffraction grating.



Top view. Measure x (centre to the first-order spot, best found as half the distance between the two first-order spots) and D (grating to screen). Then tan θ = x/D and nλ = d sin θ.

CORE PRACTICAL 6: METHOD IN BRIEF

1. Clamp the grating so it is **perpendicular to the laser beam** and **parallel to the screen** (a wall), about 1–2 m away.
2. Measure D, the perpendicular distance from grating to screen, with a metre rule or tape.
3. Mark the central spot and the spots of each order on paper on the screen.
4. Measure the distance **between the two first-order spots** (left to right) and halve it to get x. This halves the percentage uncertainty compared with measuring one side.
5. $\theta = \tan^{-1}(x/D)$. Then $\lambda = d \sin \theta / n$, with d from the grating's lines per mm.
6. Repeat for the second and third orders. Plot **sin θ against n**: a straight line through the origin with **gradient λ/d**.

52 Safety

- Use a low-power (class 2) laser. **Never look into the beam** or its reflections.
- Point it away from people (and their eye level), stand behind it, and remove shiny surfaces from the beam's path.

52 Errors and improvements

- Make D **large** and measure across **both** sides of the pattern: the same ±1 mm is then a smaller percentage.
- Measure to the **centre** of each spot.
- **Don't use the small-angle approximation.** θ is often 10°–30°, where sin θ and tan θ are noticeably different.
- The distance from the laser to the grating doesn't matter.

WORKED EXAMPLE: Λ FROM A SCREEN MEASUREMENT

Grating 500 lines per mm, so d = 1 ÷ (5.00 × 10^5) = 2.00 × 10^{-6} m. D = 1.200 m. The two first-order spots are 79.0 cm apart.
x = 0.790 ÷ 2 = 0.395 m. tan θ = 0.395 ÷ 1.200 = 0.329 → θ = 18.2°.
λ = d sin θ / n = 2.00 × 10^{-6} × sin 18.2° ÷ 1 = **6.25 × 10^{-7} m (625 nm)** (3 s.f.).
Using x/D instead of sin θ would give 658 nm: about 5 % too high.

MARK-SCHEME LANGUAGE

- Measurements: "distance from the central maximum to another maximum (or between two maxima of the same order)" **and** "the perpendicular distance from grating to screen" (WPH12 Oct 2024).
- Safety: "do not look directly into the laser / point it away from people / stand behind it / avoid reflective surfaces".
- Reduce uncertainty: "measure from the first order on one side to the first order on the other", "increase the grating-to-screen distance", so the percentage uncertainty falls (WPH12 Jan 2021).

Often asked as: what to measure, laser safety, reducing uncertainty, set-up MCQs (grating parallel to the screen; laser-to-grating distance not needed), then the 51 calculations.

SPEC 53

Electron diffraction: evidence that electrons behave as waves

WHAT THIS SPEC POINT ASKS

Understand how electron diffraction experiments give evidence that electrons have a wave nature.

53 The experiment

1. In an evacuated **electron diffraction tube**, electrons from a heated cathode are **accelerated** through a p.d. of a few kV.
2. The beam passes through a very thin film of **polycrystalline graphite**.
3. On the fluorescent screen you see **concentric rings**, not a single spot.

53 Why it is evidence of wave behaviour

- Rings like these are a **diffraction / interference pattern**: the same thing light does through a grating. Particles alone would give one spot (or a blur).
- The regular layers of carbon atoms act like a **grating**. Diffraction is strong because the electrons' **de Broglie wavelength** (54) is **similar to the spacing between the atoms** (about 10^{-10} m).
- **Raise the accelerating p.d.** → electrons go faster → greater momentum → **shorter wavelength** (λ = h/p) → less diffraction → **rings get smaller**.
- The same electrons hit the screen at single points, like particles. They show both behaviours.

WORKED EXAMPLE: WHY AN ORDINARY GRATING WON'T DO

Electrons accelerated through 2500 V have λ = 2.46 × 10^{-11} m (see 54). A light grating with 500 lines per mm has d = 2.00 × 10^{-6} m.
sin θ = λ/d = 2.46 × 10^{-11} ÷ 2.00 × 10^{-6} = 1.2 × 10^{-5} → θ ≈ 7 × 10^{-4}°.
Far too small to see. The gap must be comparable to λ, which is why atomic spacings are used.

MARK-SCHEME LANGUAGE

- "Electrons **diffract** as they pass through the graphite", so "electrons **can behave as / show the properties of** waves".
- "The (de Broglie) wavelength of the electrons is **similar to the spacing** between the atoms"; graphite has a "regular/ordered structure" acting as a grating.
- Higher p.d.: "greater momentum, so λ = h/p is smaller, so the rings get smaller".

Often asked as: interpret ring patterns (2–4 marks), effect of changing the accelerating p.d., MCQs on what counts as evidence of wave behaviour, why a neutron or an ordinary grating wouldn't work.

COMMON MISTAKE

Writing "electrons **are** waves". They **behave as** waves here (WPH12 Oct 2024 examiner report).
Saying "wave behaviour" without naming **diffraction** or interference.
Thinking attraction between charges, or electrons hitting the screen as dots, is evidence of wave nature (it isn't).

BEYOND THE SPEC, BUT EXAMINED

Is a particle suitable for diffraction by atoms? Compare its wavelength with the atomic spacing. A neutron with $\lambda = 2.8 \times 10^{-14}$ m is far smaller than an atom (about 10^{-10} m), so it would not diffract noticeably. The final mark is the numerical comparison (WPH12 Oct 2020).

SPEC 54

The de Broglie equation

WHAT THIS SPEC POINT ASKS

Use λ = h/p for the wavelength of a moving particle.

54 The equation

$$\lambda = h / p = h / mv$$

λ = de Broglie wavelength (m), h = Planck constant = 6.63×10^{-34} J s, p = momentum (kg m s^{-1}) = mass (kg) × velocity (m s^{-1}). On the formulae sheet.

- Every moving particle has a wavelength. Bigger momentum → shorter wavelength.
- Everyday objects have huge momentum compared with h, so their wavelengths are absurdly small and no wave behaviour is ever seen.
- For an electron accelerated from rest through p.d. V: kinetic energy gained = eV = ½mv^2, so find v, then p = mv, then λ.

WORKED EXAMPLE: ELECTRON AND BALL

(a) An electron ($m = 9.11 \times 10^{-31}$ kg) is accelerated from rest through 2500 V.
$E_k = eV = 1.60 \times 10^{-19} \times 2500 = 4.00 \times 10^{-16}$ J.
$v = \sqrt{(2E_k/m)} = \sqrt{(2 \times 4.00 \times 10^{-16} \div 9.11 \times 10^{-31})} = 2.96 \times 10^{7}$ m s^{-1}.
$p = mv = 9.11 \times 10^{-31} \times 2.96 \times 10^{7} = 2.70 \times 10^{-23}$ kg m s^{-1}.
$\lambda = h/p = 6.63 \times 10^{-34} \div 2.70 \times 10^{-23} =$ **2.46×10^{-11} m** (3 s.f.). Similar in size to atomic spacings, so it diffracts.
(b) A 0.058 kg tennis ball at 50 m s^{-1}: $\lambda = 6.63 \times 10^{-34} \div (0.058 \times 50) =$ **2.3×10^{-34} m**. Far too small to show any diffraction.

MARK-SCHEME LANGUAGE

- "Use of p = mv", "use of λ = h/p"; with a p.d., "use of eV = ½mv^2".
- Reverse questions: find m from λ and v, then say what the particle is (9.1 × 10^{-31} kg → electron; "similar" accepted, WPH12 Jan 2023).
- Some parts require "at least 3 s.f." in a show-that (WPH12 Jan 2026).

Often asked as: direct λ calculations, electrons through a p.d., rearrangement MCQs (v = h/mλ), a car or person's wavelength, find the particle from its mass.

COMMON MISTAKE

Dividing h by the **velocity** instead of the momentum (scored 0, WPH12 Jan 2025 examiner report).
Putting the electron **charge** in where the **mass** belongs.
Leaving off the conclusion when the question ends with a claim to check.

BEYOND THE SPEC, BUT EXAMINED

- Momentum straight from kinetic energy: E_k = p^2/2m, so p = √(2mE_k) (accepted, WPH12 Jan 2026).
- **Photons have momentum** too: p = h/λ (WPH12 Jan 2021, MCQ).

SPEC 55

Transmission and reflection at an interface

WHAT THIS SPEC POINT ASKS

Understand that waves can be transmitted and reflected where two media meet.

55 What happens at a boundary

- When a wave reaches the boundary between two media in which it travels at **different speeds** (different densities or materials), part of it is **reflected** and part is **transmitted** (refracted if it arrives at an angle). Some energy may also be **absorbed**.
- The **bigger the difference** between the two media, the **more is reflected** and the less is transmitted. Similar media → mostly transmitted.
- Reflection obeys the law of reflection: angle of incidence = angle of reflection (measured from the normal).
- As a wave travels on through a medium, absorption keeps reducing its amplitude, so echoes from deep boundaries are weak.

55 Examples

Boundary	What happens	Consequence
air–skin (ultrasound)	very different: almost all reflected	**gel** is used to remove the air layer so the ultrasound gets in
soft tissue–soft tissue	similar: small reflection, most transmitted	each organ boundary gives a small echo, and deeper ones can still be reached
soft tissue–bone or lung (air)	very different: most reflected	little reaches what lies behind
steel–air in a cracked rail	very different	a crack gives a strong echo

MARK-SCHEME LANGUAGE

- "Reflects at a **boundary between different materials / media / densities**"; "the speed of ultrasound in air is different from that in steel" is accepted.
- "**Some** of the pulse is transmitted / passes through."
- "Bounces back" is ignored. Write **reflected**.

Often asked as: why part of a pulse is reflected, why deeper features give weaker or no echoes, why little ultrasound reaches the lungs (MCQ).

BEYOND THE SPEC, BUT EXAMINED

If the first crack reflects most of the pulse, too little energy reaches a deeper crack for its echo to be detected (WPH12 Oct 2019).

SPEC 56

Pulse-echo techniques

WHAT THIS SPEC POINT ASKS

Understand how a pulse-echo technique locates an object, and how the wavelength and the pulse duration limit the information you can get.

56 How it works

1. A transmitter sends out a **short pulse** of waves (ultrasound, sonar, radar, laser light).
2. The pulse is **reflected** from each boundary or object it meets.
3. A receiver detects the echoes and the **time** between sending and receiving is measured.
4. The pulse goes there **and back**, so with the wave speed known:

$$\text{distance} = v\,t\,/\,2$$

The time of each echo gives the depth of each boundary; the strength of the echo says something about the boundary.

56 What limits the detail

Limit	Why	Consequence
Wavelength	waves diffract round objects about the size of λ or smaller, so they don't reflect a clear echo	you can't resolve detail smaller than about λ. **Higher frequency → shorter λ → finer detail**. But high frequencies are absorbed more, so they don't reach as deep
Pulse duration	a long pulse means a long echo; echoes from two close boundaries overlap	two boundaries closer than about (v × pulse duration)/2 can't be told apart. **Shorter pulses → better depth resolution**
Time between pulses	each echo must return before the next pulse is sent, or you can't tell which pulse it came from	maximum range = v × (time between pulses)/2

WORKED EXAMPLE: ULTRASOUND SCAN

Ultrasound in soft tissue travels at 1540 m s^{-1}.
(a) An echo returns 65 μs after the pulse: depth = vt/2 = 1540 × 65 × 10^{-6} ÷ 2 = **0.050 m (5.0 cm)**.
(b) At 3.5 MHz, λ = v/f = 1540 ÷ 3.5 × 10^{6} = 4.4 × 10^{-4} m, so detail smaller than about **0.4 mm** can't be resolved.
(c) A 2.0 μs pulse is 1540 × 2.0 × 10^{-6} = 3.1 mm long in tissue. Boundaries closer than about half this, **1.5 mm**, give overlapping echoes.
(d) Pulses sent every 0.20 ms: the deepest echo that returns in time comes from 1540 × 0.20 × 10^{-3} ÷ 2 = **0.15 m**.

MARK-SCHEME LANGUAGE

- 6-mark structure: pulses sent; **reflected** at boundaries; **time** between sending and receiving measured; speed known; distance = speed × time; **half** the time (or half the distance).
- "Uses half time or half distance" is almost always a separate mark.
- Resolution: "higher frequency, shorter wavelength", "more detail / better resolution", "smaller objects can be detected".

Often asked as: timing calculations with the factor of 2 (often ending in a claim to check), 6-mark descriptions of the technique, why a higher frequency is used, the effect of pulse length or the time between pulses.

COMMON MISTAKE

Forgetting to halve the time.
"Bounces back" instead of "reflected" (WPH12 Jan 2020 examiner report).
Not converting ms or μs to seconds (WPH12 Jun 2024 examiner report).
Saying a longer gap between pulses "increases distance" without the reason: the echo must return before the next pulse goes out (WPH12 Jun 2024).

BEYOND THE SPEC, BUT EXAMINED

Laser altimetry from a satellite: height = c × t/2 (e.g. 3.20 ms gives 480 km); only photons from the laser should be counted, as other light adds false signals (WPH12 Oct 2021).

SPEC 57

The wave model and the photon model of light

WHAT THIS SPEC POINT ASKS

Understand how electromagnetic radiation can be described by a wave model and by a photon model, and how these models developed over time.

57 Two models, each needed

Behaviour of light	Explained by
Reflection, refraction	both models (the wave model correctly predicts light slows in glass)
Diffraction, interference, polarisation	**wave model** only
Photoelectric effect (59–62), line spectra (63)	**photon model** only

- **Wave model:** light is a transverse electromagnetic wave. Its energy is spread continuously over the wavefront, and the energy delivered depends on **intensity** (amplitude squared) and time.
- **Photon model:** light travels as **photons**, discrete packets (quanta) of energy E = hf. Each photon interacts with **one** electron and gives up **all** its energy at once.
- Light needs both models: this is **wave–particle duality**. Electrons show it too (53).

57 How the ideas developed

When	Who	Step
late 1600s	Newton	light as a stream of **particles (corpuscles)**; he wrongly predicted light speeds up in glass
late 1600s	Huygens	light as a **wave**, using secondary wavelets
1801	Young	**two-slit interference**: strong evidence for waves; wave model accepted
1850	Foucault	light measured to be **slower** in water: agrees with waves, not Newton
1860s–1880s	Maxwell, then Hertz	light is an **electromagnetic wave**; Hertz makes radio waves and notices that UV helps sparks jump (the photoelectric effect)
1900	Planck	energy is emitted in **quanta**, E = hf
1905	Einstein	light itself comes in **photons**: explains the photoelectric effect
1916	Millikan	experiments confirm Einstein's photoelectric equation
1924–1927	de Broglie; Davisson and Germer, G. P. Thomson	particles have wavelengths; **electron diffraction** confirms it

A model is kept while it explains the evidence. When a new observation (like the photoelectric effect) can't be explained, the model is extended or a new one is added.

MARK-SCHEME LANGUAGE

The marks are for **what each model predicts**, not dates:
- Wave model: energy builds up over time, so emission would be **delayed** in dim light; **any frequency** would work given enough intensity or time; electron energy would depend on **intensity**.
- Photon model: a **minimum (threshold) frequency** is needed; emission is **instantaneous**; kinetic energy depends on **frequency**; intensity changes the **number** of photons, not their energy.

Often asked as: MCQ on which model explains which effect; 6-mark comparisons of the two models using the photoelectric effect (see 62).

BEYOND THE SPEC, BUT EXAMINED

"In the wave model, more intense light means a bigger amplitude, so any intensity would eventually release electrons. In the photon model, intensity changes the number of photons, not their energy, so only UV photons with enough energy release electrons" (5 marks, WPH12 Jan 2025).

SPEC 58 Photon energy, E = hf

WHAT THIS SPEC POINT ASKS

Use E = hf to link the energy of a photon to the frequency of the radiation.

58 The equation

A **photon** is a discrete packet (quantum) of electromagnetic energy. It has no mass and no charge.

$$E = h f = h c / \lambda$$

E = photon energy (J), h = 6.63 × 10^{-34} J s, f = frequency (Hz), c = 3.00 × 10^{8} m s^{-1}, λ = wavelength (m). E = hf is on the formulae sheet.

- Higher frequency (shorter wavelength) → more energy per photon. Gamma and X-ray photons carry the most; radio photons the least.
- **Photons per second** from a source of power P: N = P / E = P / hf. For a beam, P = intensity × area (44).
- For the same power, shorter-wavelength light means fewer photons per second, because each carries more energy.

WORKED EXAMPLE: PHOTON ENERGY AND PHOTON RATE

(a) Blue light, λ = 450 nm: E = hc/λ = (6.63 × 10^{-34} × 3.00 × 10^{8}) ÷ (450 × 10^{-9}) = **4.42 × 10^{-19} J** (3 s.f.) = 2.76 eV.
(b) A 5.0 mW laser emits 633 nm light. E per photon = 1.99 × 10^{-25} ÷ 633 × 10^{-9} = 3.14 × 10^{-19} J.
Photons per second = P/E = 5.0 × 10^{-3} ÷ 3.14 × 10^{-19} = **1.6 × 10^{16} s^{-1}** (2 s.f.).

MARK-SCHEME LANGUAGE

- "Use of v = fλ" (or c = fλ) and "use of E = hf". A single combined substitution into E = hc/λ earns both marks.
- Photon counting: "E = Pt", then "number = total energy ÷ energy of one photon".

Often asked as: on nearly every paper: E from λ (and back), photons emitted per second or per minute, percentage of photons that release an electron.

COMMON MISTAKE

Forgetting nm → m (× 10^{-9}).
In a "show that", giving only the rounded value from the question. Show more figures than it gives (e.g. 4.43 eV for "4.4 eV").

BEYOND THE SPEC, BUT EXAMINED

Photocell efficiency: electrons per second = current ÷ e (I/e), compared with photons per second arriving, e.g. 82 % of photons release an electron (WPH12 Jan 2026).

SPEC 59 Absorbing a photon can release a photoelectron

WHAT THIS SPEC POINT ASKS

Understand that when a photon is absorbed, an electron (a photoelectron) can be emitted.

59 The photoelectric effect

When electromagnetic radiation of high enough frequency shines on a metal surface, **electrons are emitted** from the surface. These are called **photoelectrons**.

59 What happens, step by step

1. A **photon** hits the metal and is **absorbed by one electron**: a **one-to-one** interaction. The photon gives up **all** its energy; it can't be partly absorbed.
2. Part of that energy is needed to free the electron from the metal (at least the **work function**, 60).
3. If the photon's energy is enough, the electron escapes. Whatever energy is left over becomes its **kinetic energy**.
4. Electrons from deeper inside lose some energy on the way out, so photoelectrons leave with a **range** of kinetic energies, up to a **maximum**.
5. If the photon doesn't have enough energy, no electron escapes, however many photons arrive. An electron can't save up energy from two photons.

59 Intensity and frequency

- **Brighter light** (same frequency) = **more photons per second** → more electrons emitted per second → bigger photocurrent. The maximum kinetic energy is unchanged.
- **Higher frequency** = more energy per photon → **faster** photoelectrons (more kinetic energy). The number per second depends on how many photons arrive.

59 The gold-leaf electroscope demonstration

- A clean **zinc** plate on a **negatively** charged electroscope: the leaf stands up.
- Shine **UV** on it: the leaf **falls** at once. Photoelectrons leave, so the plate loses its negative charge.
- Shine bright **visible** light: nothing happens. Its photons have too little energy for zinc.
- Charge the plate **positively**: UV makes no difference, because any released electrons are pulled back.

MARK-SCHEME LANGUAGE

- "A photon is absorbed by an electron"; "**one photon interacts with one electron**"; "the electron gains the photon's energy and is emitted / escapes".
- "Energy is shared between the work function and the kinetic energy of the electron".
- Intensity: "more photons **per second**, so more electrons **per second**, so a greater current". The idea of a rate is needed.

Often asked as: describe the process (2–4 marks), why photoelectrons have a range of speeds, the effect of intensity on the current, MCQs (a photon can't be partly absorbed).

COMMON MISTAKE

Writing answers where it isn't clear whether photons or electrons are being released (WPH12 Oct 2020 examiner report). Photons go **in**; electrons come **out**.
Saying brighter light gives faster electrons.

BEYOND THE SPEC, BUT EXAMINED

A photocell has a metal cathode and a collecting gauze. With the gauze positive, electrons are attracted across and a current flows; reverse the polarity and the current stops once the p.d. is large enough (WPH12 Oct 2023).

SPEC 60 Threshold frequency, work function and hf = φ + ½mv²max

WHAT THIS SPEC POINT ASKS

Know what threshold frequency and work function mean, and use hf = φ + ½mv²max.

60 Definitions

- **Work function φ**: the **minimum** energy needed to remove an electron from the **surface** of a metal. Each metal has its own value.
- **Threshold frequency f_0**: the **minimum** frequency of radiation that will release electrons from the metal. A photon at f_0 has just enough energy: **$hf_0 = \varphi$**, so $f_0 = \varphi/h$. The matching threshold wavelength is $\lambda_0 = c/f_0$ (longer wavelengths won't work).

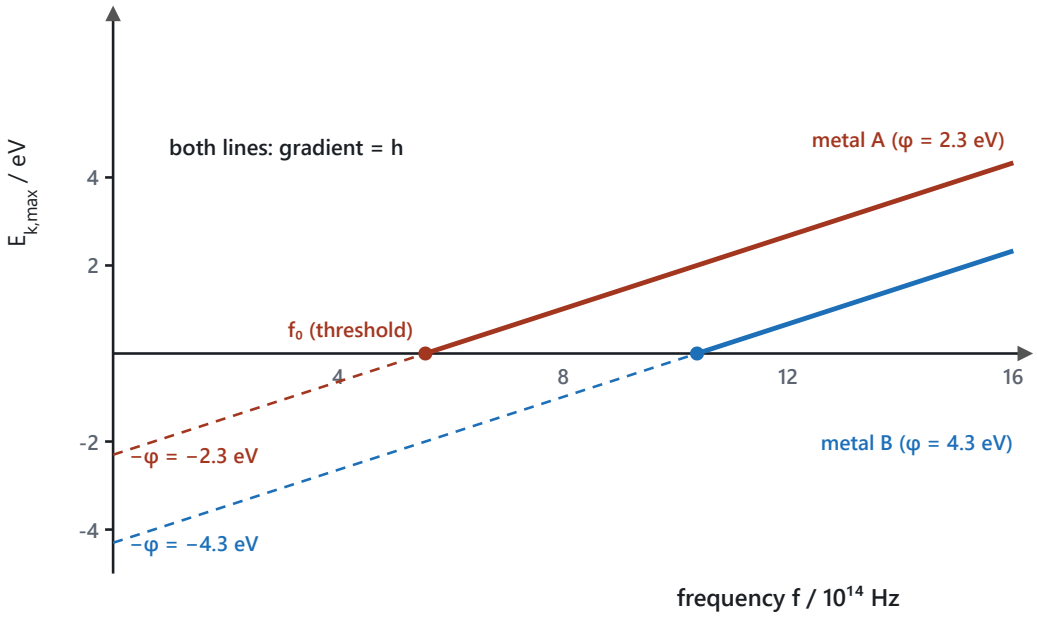
60 Einstein's photoelectric equation

Energy conservation for the most energetic electrons (those from the surface):

$$h f = \varphi + \tfrac{1}{2} m v^2_{max}$$

hf = photon energy (J), φ = work function (J), ½mv²max = maximum kinetic energy of a photoelectron (J), m = 9.11 × 10^{-31} kg. On the formulae sheet.

60 The graph



E_k,max = hf − φ. Each metal gives a straight line of gradient h. It meets the f-axis at the threshold frequency f_0 and, extended back, the y-axis at −φ. A metal with a bigger φ gives a parallel line further right.

60 Stopping potential

Make the collector negative and increase the p.d. until the current just stops. Then even the fastest electrons are turned back, so the work done equals their kinetic energy:

$$e V_s = \tfrac{1}{2} m v^2_{max}$$

WORKED EXAMPLE: ZINC IN ULTRAVIOLET

Zinc has φ = 4.3 eV. UV of wavelength 200 nm shines on it.
Photon energy: $E = hc/\lambda = 1.99 \times 10^{-25} \div 200 \times 10^{-9} = 9.95 \times 10^{-19}$ J.
φ in joules: $4.3 \times 1.60 \times 10^{-19} = 6.88 \times 10^{-19}$ J.
$E_{k,max} = hf - \varphi = 9.95 \times 10^{-19} - 6.88 \times 10^{-19} = 3.07 \times 10^{-19}$ J.
$v_{max} = \sqrt{(2E_k/m)} = \sqrt{(2 \times 3.07 \times 10^{-19} \div 9.11 \times 10^{-31})} =$ **8.2×10^5 m s^{-1}** (2 s.f.).
Stopping potential: $V_s = E_{k,max}/e = 3.07 \times 10^{-19} \div 1.60 \times 10^{-19} =$ **1.9 V**.
Threshold frequency: $f_0 = \varphi/h = 6.88 \times 10^{-19} \div 6.63 \times 10^{-34} =$ **1.0×10^{15} Hz** (threshold wavelength 290 nm). Visible light (400–700 nm) can't release electrons from zinc.

MARK-SCHEME LANGUAGE

- Work function: "**minimum** energy required to release/remove an electron from the **surface** of a metal". "Energy needed" without "minimum" is ignored.
- "Converts φ from eV to J" is a marking point. "Use of $hf = \varphi + \frac{1}{2}mv^2_{max}$."
- The speed mark needs real energy and mass values substituted (not the charge).
- Graph: "gradient = h", "x-intercept = threshold frequency", "y-intercept = −φ".

Often asked as: define φ, find v_{max}, identify a metal from φ in a table, decide whether emission happens, threshold frequency, stopping potential, read h and φ from a graph.

COMMON MISTAKE

Forgetting "minimum" in the work function definition (WPH12 Oct 2020 examiner report).
Mixing eV and J in one subtraction.
Saying brighter light increases the electrons' speed. Only frequency does (WPH12 Jan 2021).
No unit on v_{max} or V_s.

BEYOND THE SPEC, BUT EXAMINED

- A photoelectron then **accelerated** through a p.d. V gains eV, so final E_k = initial E_k + eV, e.g. 8.30×10^{-19} J + 1.73×10^{-17} J = 1.81×10^{-17} J (WPH12 Oct 2024).
- One paper (WPH12 Jun 2022) gave a graph with deliberately wrong intercepts and asked you to show it can't be zinc. Don't learn its −10 eV intercept as data.

SPEC 61

The electronvolt

WHAT THIS SPEC POINT ASKS

Use the electronvolt (eV) to express small energies.

61 Definition

1 electronvolt is the energy transferred when **one electron** moves through a **potential difference of 1 volt**. From W = QV:

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ C} \times 1 \text{ V} = 1.60 \times 10^{-19} \text{ J}$$

- **eV → J:** multiply by 1.60×10^{-19}. **J → eV:** divide by 1.60×10^{-19}.
- An electron accelerated from rest through V volts gains **V electronvolts** of kinetic energy.

- Photon energies, work functions and atomic energy levels are a few eV, so the eV keeps the numbers simple.

WORKED EXAMPLE: CONVERSIONS

(a) An electron accelerated from rest through 150 V gains 150 eV = $150 \times 1.60 \times 10^{-19} = 2.40 \times 10^{-17}$ J. Its speed: $v = \sqrt{(2 \times 2.40 \times 10^{-17} \div 9.11 \times 10^{-31})} = \mathbf{7.26 \times 10^{6}\ m\ s^{-1}}$ (3 s.f.).
(b) A 700 nm photon: $E = 1.99 \times 10^{-25} \div 700 \times 10^{-9} = 2.84 \times 10^{-19}$ J $= 2.84 \times 10^{-19} \div 1.60 \times 10^{-19} =$ **1.78 eV**.

MARK-SCHEME LANGUAGE

- "Converts J to eV" / "eV to J" is usually its own mark.
- In a comparison either unit is fine if both values are in the same unit (e.g. 0.66 eV < 1.5 eV, WPH12 Jan 2025).

Often asked as: a one-mark conversion inside photoelectric and energy-level questions.

COMMON MISTAKE

Treating the eV as a unit of voltage. It is a unit of **energy**.
Comparing a value in eV with one in J.

SPEC 62 The photoelectric effect as evidence for the particle nature of light

WHAT THIS SPEC POINT ASKS

Understand how the photoelectric effect gives evidence that electromagnetic radiation behaves as particles.

62 Observation, wave prediction, photon explanation

Observation	What the wave model predicts	Photon explanation
No electrons below a **threshold frequency**, however bright the light	any frequency should work if the light is intense enough (or shone for long enough)	each photon has energy hf; below f_0 one photon has less than φ, and one electron can't combine energy from several photons
Emission is **instantaneous**, even in very dim light	in dim light an electron would have to absorb energy for a long time before it could escape	one photon gives all its energy to one electron at once
Maximum kinetic energy depends on frequency, not intensity	more intense light (bigger amplitude) should give faster electrons	$E_{k,max} = hf - \varphi$: only frequency changes the photon energy
Number of electrons per second depends on **intensity**	(not a clear prediction)	more intense = more photons per second = more electrons per second

62 Writing the 6-mark answer

1. Light arrives as **photons**, each with energy **E = hf**.
2. **One photon** interacts with **one electron**.
3. An electron needs a **minimum energy (the work function)** to escape, so there's a **threshold frequency**.
4. So emission is **instantaneous** when $f > f_0$; in the wave model energy would have to build up.

5. Extra energy becomes kinetic energy, so **frequency** controls E_k,max; **intensity** only controls the **number** of electrons per second.
6. These can only be explained if light behaves as **particles**.

MARK-SCHEME LANGUAGE

Indicative content credited (WPH12 Jan 2022): photon energy related to frequency; one-to-one interaction; minimum energy/frequency needed; instantaneous release; frequency affects kinetic energy; intensity affects the number of electrons per second. Each "observation" needs its "explanation" for the linkage marks.

Often asked as: 6-mark questions (WPH12 Jun 2019, Jan 2022, Oct 2023), 5-mark wave v photon comparisons, why UV releases electrons but visible light doesn't, two sources and two metals.

COMMON MISTAKE

Saying "brighter light gives more energetic electrons".
Describing the observations without saying why the **wave** model fails, when the question asks you to compare.

BEYOND THE SPEC, BUT EXAMINED

A wave-model estimate of the delay: a single atom (area about 8×10^{-20} m^2) in dim light would take about two minutes (118 s) to collect enough energy, yet emission is instant (WPH12 Oct 2020).

SPEC 63

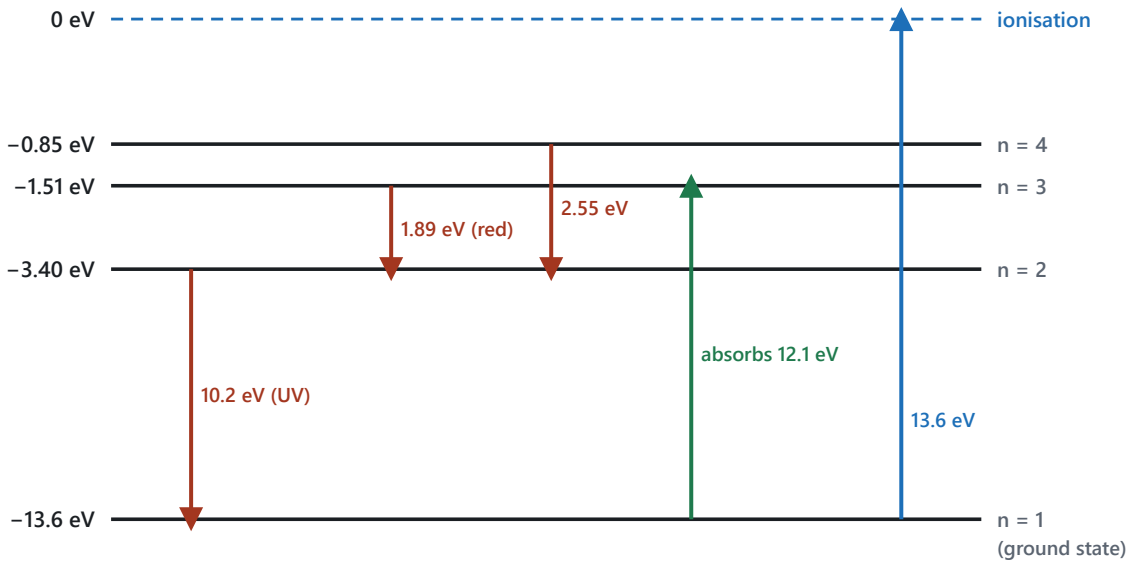
Line spectra and energy levels

WHAT THIS SPEC POINT ASKS

Explain atomic line spectra using transitions between discrete energy levels, and calculate the frequency of the radiation emitted or absorbed in a transition.

63 Discrete energy levels

- Electrons in an atom can only have certain **discrete (fixed) energies**: the **energy levels**. Each element has its own set.
- The levels are **negative**: 0 eV is the energy of an electron that has just been removed (**ionisation**). The lowest level is the **ground state**.
- **Ionisation energy** = the energy needed to remove an electron from the ground state completely (13.6 eV for hydrogen).



Hydrogen. Emission arrows point down (photon out), absorption arrows point up (photon in). The bigger the jump, the bigger the photon energy and the shorter the wavelength.

63 Emission spectra

1. Atoms gain energy (in a discharge tube, collisions with fast electrons; or by heating).
2. An electron is **excited** to a higher energy level.
3. It is unstable there and drops back to a lower level, **emitting a photon** whose energy equals the **difference** between the two levels.
4. The levels are discrete, so only certain energy differences exist, so only **certain frequencies (wavelengths)** are emitted: a **line spectrum** of bright lines on a dark background.

$$h f = E_1 - E_2$$

E_1 = upper level, E_2 = lower level (J; convert from eV). f = frequency of the photon (Hz).

63 Absorption spectra

- Shine white light (all wavelengths) through a cool gas. Photons with **exactly** the energy of an upward transition are absorbed; all others pass straight through.
- You see a continuous spectrum crossed by **dark lines** at the **same** wavelengths as that gas's emission lines.
- A photon with the wrong energy is not absorbed at all, even if it has more than enough. (An electron colliding with the atom is different: it can hand over just part of its kinetic energy.)

WORKED EXAMPLE: HYDROGEN TRANSITIONS

(a) n = 3 (−1.51 eV) to n = 2 (−3.40 eV): $\Delta E = 3.40 - 1.51 = 1.89$ eV $= 1.89 \times 1.60 \times 10^{-19} = 3.02 \times 10^{-19}$ J. $f = \Delta E/h = 3.02 \times 10^{-19} \div 6.63 \times 10^{-34} =$ **4.56×10^{14} Hz**; $\lambda = c/f =$ **6.58×10^{-7} m (658 nm, red)**.

(b) n = 2 to n = 1: $\Delta E = 13.6 - 3.40 = 10.2$ eV $= 1.63 \times 10^{-18}$ J, $f = 2.46 \times 10^{15}$ Hz, $\lambda =$ **1.22×10^{-7} m** (ultraviolet).

(c) Can a 12.3 eV photon be absorbed by hydrogen in the ground state? −13.6 + 12.3 = −1.3 eV, which is not a level (they are −1.51 and −0.85). So **no**: the photon passes through and the electron stays in the ground state.

63 How many lines?

An electron in level n can come down in steps or in one jump. From n = 4 the possible jumps are 4→3, 4→2, 4→1, 3→2, 3→1, 2→1: **six** lines.

MARK-SCHEME LANGUAGE

- "**Electrons** gain energy / are excited", "move to **higher** energy levels", "fall to **lower** levels", "**emitting photons**", "energy levels are **discrete**", "photon energy = **difference** between levels", "E = hf so only certain frequencies".
- Not "atoms" for electrons in some schemes (WPH12 Jun 2019); "gain energy" alone wasn't enough for the first point in another (WPH12 Oct 2021).
- Absorption: "a photon can only excite an electron if its energy **matches** the difference between levels".

Often asked as: 6-mark explanations of a hydrogen or discharge-tube spectrum, λ or f from a transition, which transition gives a given photon, is a given photon absorbed, MCQs on allowed jumps and highest frequency.

COMMON MISTAKE

Thinking a bigger jump gives a longer wavelength. Bigger jump → bigger energy → **shorter** wavelength (WPH12 Oct 2022 examiner report).
Assuming an electron in n = 3 can only drop to n = 2 or n = 1 in one go. Count every possible jump, including n = 3 → 2 → 1 (WPH12 Jan 2023 examiner report).
Talking about the work function in an energy-level question. That belongs to metals and the photoelectric effect (WPH12 Jan 2021 examiner report).
Quoting a transition the wrong way round: emission goes **down**, absorption goes **up**.

BEYOND THE SPEC, BUT EXAMINED

- Air glows in several colours because it contains different gases, each with its own energy levels (WPH12 Oct 2021).
- Mercury lamps: raising the p.d. doesn't change the UV frequency because the energy levels are fixed (WPH12 Jun 2025).
- Equations for the Bohr atom may be given in the stem; just substitute (e.g. ground-state energy 13.56 eV, WPH12 Oct 2020).

Topic: Electric Circuits

SPEC 64

Electric current

WHAT THIS SPEC POINT ASKS

Know that current is the rate of flow of charged particles, and use $I = \Delta Q/\Delta t$.

64 What current is

- An electric current is a flow of **charge carriers**. In metals the carriers are **free (conduction) electrons**. In electrolytes and ionised gases they are **ions** (positive and negative).
- **Current** is the **rate of flow of charge**:

$$I = \Delta Q / \Delta t$$

- I = current (ampere, A); ΔQ = charge passing a point (coulomb, C); Δt = time taken (s).
- So **1 C = 1 A s**: one coulomb is the charge that passes when a current of 1 A flows for 1 s.
- Charge comes in whole numbers of the electron charge $e = 1.60 \times 10^{-19}$ C (given in the exam). The number of electrons carrying a charge Q is **N = Q/e**. One coulomb is about 6.25×10^{18} electrons.

64 Direction and measurement

- **Conventional current** is drawn from + to − round the outside of the circuit. Electrons actually drift the other way (− to +). Both pictures give the same answers as long as you stay consistent.
- Measure current with an **ammeter in series**. An ideal ammeter has **zero resistance**, so it doesn't change the current it measures.

WORKED EXAMPLE: CHARGE AND NUMBER OF ELECTRONS

A phone charger supplies a steady 1.8 A for 25 minutes. How much charge flows, and how many electrons is that?

t = 25 × 60 = 1500 s (convert minutes first)

Q = It = 1.8 × 1500 = **2700 C**

N = Q/e = 2700 / 1.60×10^{-19} = 1.69×10^{22} ≈ **1.7×10^{22} electrons** (a number: no unit)

Electrons per second at 0.64 A: 0.64 / 1.60×10^{-19} = **4.0×10^{18} s^{-1}**.

MARK-SCHEME LANGUAGE

- "Use of I = ΔQ/Δt" (or Q = It) is the first mark. A count of electrons needs **no unit**.
- Divide the charge by e to get a number of electrons; never multiply by e.

Often asked as: number of electrons passing in a time or per second, time for a given charge, unit and base-unit MCQs, the meaning of symbols in I = nqvA.

COMMON MISTAKE

Forgetting to convert minutes or hours to seconds. Multiplying by e instead of dividing.
Using a value from an earlier part of the question that the new part doesn't need (Oct 2019 examiner report: a time from part (i) was carried into part (ii) by mistake).

BEYOND THE SPEC, BUT EXAMINED

Base units of electrical quantities turn up in MCQs. Build them from equations you know (WPH12 Jun 2021):
- coulomb = **A s** (from Q = It)
- joule = **kg m^2 s^{-2}** (from W = Fs); watt = **kg m^2 s^{-3}** (from P = W/t)
- volt = **kg m^2 s^{-3} A^{-1}** (from V = P/I); ohm = **kg m^2 s^{-3} A^{-2}** (from R = V/I)

The current is a **base** quantity; power, charge and p.d. are **derived** (WPH12 Oct 2020).

SPEC 65

Potential difference

WHAT THIS SPEC POINT ASKS

Use V = W/Q: the potential difference between two points is the energy transferred per unit charge.

65 Energy per coulomb

- The **potential difference (p.d.)** between two points is the **work done (energy transferred) per unit charge** as charge moves between them. In a component, electrical energy is transferred to other stores (thermal, light, kinetic).

$$V = W / Q$$

- V = p.d. (volt, V); W = energy transferred (J); Q = charge (C). So **1 V = 1 J C^{-1}**.
- Combine with Q = It to get the energy transferred in a time: **W = VIt** (see 69).
- Measure p.d. with a **voltmeter in parallel** (across the component). An ideal voltmeter has **infinite resistance**, so it takes no current.
- Current flows **through** a component; p.d. is **across** it.

65 A charged particle accelerated through a p.d.

- An electron moved through a p.d. V gains energy **W = eV**. If it starts from rest, this becomes kinetic energy: **eV = ½mv^2**. This links straight to the electronvolt (61) and electron diffraction (53–54).

WORKED EXAMPLE: ENERGY FROM P.D. AND CHARGE

(a) A 12 V battery pushes 4.0 × 10^4 C through a starter motor. Energy transferred?
W = QV = 4.0 × 10^4 × 12 = **4.8 × 10^5 J**
(b) An electron is accelerated from rest through 2500 V. Find its kinetic energy and speed (m = 9.11 × 10^{-31} kg).
W = eV = 1.60 × 10^{-19} × 2500 = **4.00 × 10^{-16} J**
v = √(2W/m) = √(2 × 4.00 × 10^{-16} / 9.11 × 10^{-31}) = **3.0 × 10^7 m s^{-1}**

MARK-SCHEME LANGUAGE

- "Use of V = W/Q"; W = VIt is accepted.
- p.d. = "energy transferred **per unit charge**" or "work done per unit charge". Not "per unit current" and not "energy per second".

Often asked as: energy from p.d. and charge, energy of an electron accelerated through a p.d. (often "give your answer to at least 3 s.f."), units MCQs such as J C^{-1}.

COMMON MISTAKE

"The voltage flows through the lamp." Current flows through; p.d. is across.
Mixing up p.d. and e.m.f.: p.d. is energy transferred **from** electrical form in a component; e.m.f. is energy transferred **to** electrical form by a source (see 77).

SPEC 66

Resistance and Ohm's law

WHAT THIS SPEC POINT ASKS

Know that resistance is defined by R = V/I, and that Ohm's law (I ∝ V at constant temperature) is a special case.

66 Definition of resistance

$$R = V / I$$

- R = resistance (ohm, Ω), V = p.d. across the component (V), I = current through it (A). **1 Ω = 1 V A^{-1}**.
- This **defines** resistance and works for **every** component at **every** point on its I–V graph: the resistance at a point is **V/I at that point**.

66 Ohm's law: a special case

- **Ohm's law**: the current through a conductor is **directly proportional** to the p.d. across it, **provided the temperature (and other physical conditions) stay constant**.
- A component that obeys it is **ohmic**: its I–V graph is a **straight line through the origin**, so R is the same at every p.d. A metal wire kept at constant temperature is ohmic.
- A filament lamp, diode or thermistor is **non-ohmic**: V/I changes as the p.d. changes (see 70).

66 Reading resistance from a graph

- Straight line through the origin: R = V/I at any point, and on an I–V graph the gradient = 1/R.
- Curve: take **V/I at the point** you want. **Do not use the gradient (tangent)** of a curve: that is not the resistance in this course.
- Where the current is zero for a non-zero p.d. (e.g. a reverse-biased diode), the resistance is **infinite**.

WORKED EXAMPLE: RESISTANCE OF A LAMP AT TWO P.D.S

A filament lamp carries 0.25 A at 2.0 V and 0.55 A at 12.0 V.
At 2.0 V: R = V/I = 2.0 / 0.25 = **8.0 Ω**
At 12.0 V: R = 12.0 / 0.55 = 21.8 ≈ **22 Ω**
The resistance rises because the filament is hotter at the higher p.d. (see 79). It is non-ohmic.

MARK-SCHEME LANGUAGE

- "Resistance = p.d. / current" with V and I for the **same** point.
- Zero current at a non-zero p.d. → "resistance is infinite" (WPH12 Jan 2020 MCQ).
- LED: "resistance (almost) infinite below about 1.6 V, then it decreases" (WPH12 Oct 2024).

Often asked as: R from a point on a graph, identify which component's resistance falls with p.d., part of nearly every circuit calculation.

COMMON MISTAKE

Using the gradient of a curved I–V graph as R (or as 1/R). Use V/I at the point.
Treating R = V/I as "Ohm's law". R = V/I is a definition; Ohm's law is the statement that I ∝ V at constant temperature.
Finding the whole-circuit current from only one of the resistances (Oct 2020 examiner report).

SPEC 67

Charge and energy conservation in circuits

WHAT THIS SPEC POINT ASKS

(a) Explain current distribution as a result of charge conservation. (b) Explain p.d. distribution as a result of energy conservation.

67 (a) Current: conservation of charge

- Charge can't be created or destroyed, and it doesn't pile up at any point in a circuit.
- So at a **junction**: **total current in = total current out** (Kirchhoff's first law).
- In a **series** circuit there are no junctions, so the **current is the same at every point** (including through the cell).
- In a **parallel** circuit the current **splits** between branches and **recombines**: $I = I_1 + I_2$.

67 (b) p.d.: conservation of energy

- Each coulomb gains energy in the source (the e.m.f.) and transfers all of it to the components before it returns.
- So round any **closed loop**: **sum of e.m.f.s = sum of p.d.s** (Kirchhoff's second law).
- In **series**: the p.d.s across the components **add up** to the e.m.f. ($V = V_1 + V_2$ for an ideal cell).
- In **parallel**: each branch forms its own loop with the source, so **every branch has the same p.d.** across it.

	Series	Parallel
Current	same everywhere	splits: $I = I_1 + I_2$ (charge conservation)
p.d.	shared: $V = V_1 + V_2$ (energy conservation)	same across each branch

WORKED EXAMPLE: JUNCTIONS AND LOOPS

(a) 2.5 A flows into a junction. Two of the wires leaving it carry 0.8 A and 1.1 A. Current in the third wire?
2.5 = 0.8 + 1.1 + I, so I = **0.6 A** (out of the junction)
(b) A 12.0 V cell (negligible internal resistance) is in series with a lamp and a resistor. The lamp has 7.5 V across it.
p.d. across resistor = 12.0 − 7.5 = **4.5 V**
(c) Two lamps are now put in parallel with each other in place of the single lamp. The p.d. across the resistor is 5.0 V. p.d. across each lamp = 12.0 − 5.0 = **7.0 V** (the same for both, because they are in parallel).

MARK-SCHEME LANGUAGE

Indicative points from a 6-mark series-and-parallel question (WPH12 Jun 2022):
- current is the rate of flow of charge
- current is the same at all points in a series circuit (current in C = current in the cell)
- current into a junction = current out of it (current in C = current in A + current in B)
- p.d. is energy transferred per unit charge
- p.d. is shared between components in series (p.d. across C + p.d. across A = e.m.f.)
- p.d. is the same across components in parallel

Often asked as: a 6-mark explanation of the readings in a series-parallel circuit, "sum of e.m.f.s = sum of p.d.s" MCQs, slide-wire loop MCQs.

EXAM TIP

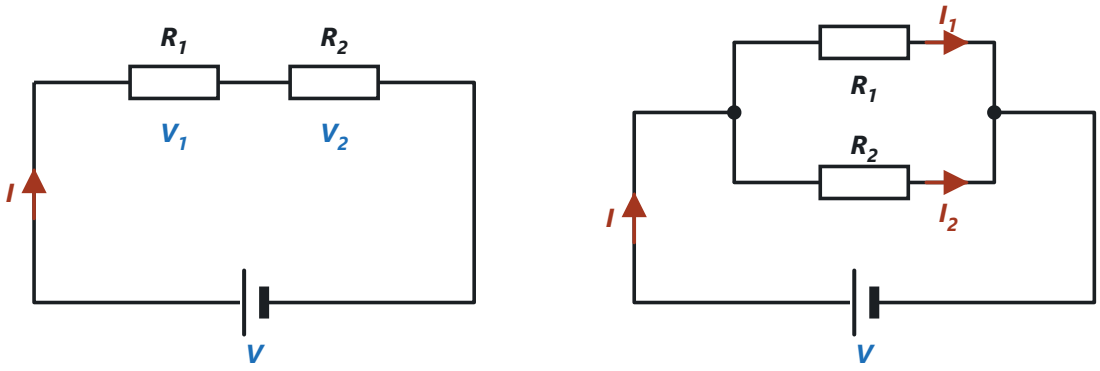
In the 6-mark version, link each rule to its conservation law in one sentence ("charge is conserved, **so** current into the junction equals current out"). The linkage marks need the "so".

SPEC 68

Resistors in series and parallel

WHAT THIS SPEC POINT ASKS

Derive the formulae for resistors in series and in parallel from charge and energy conservation, and use them.



(a) Series: same I; $V = V_1 + V_2$ **(b) Parallel: same V; $I = I_1 + I_2$**

Series: one current, p.d.s add. Parallel: one p.d., currents add.

68 Deriving the series formula

1. **Energy conservation**: the p.d.s add: $V = V_1 + V_2$.
2. **Charge conservation**: the same current I flows through both, so $V_1 = IR_1$ and $V_2 = IR_2$. For the whole combination, $V = IR$.
3. Substitute: $IR = IR_1 + IR_2$. Divide by I:

$$R = R_1 + R_2 + \ldots$$

68 Deriving the parallel formula

1. **Charge conservation**: the currents add: $I = I_1 + I_2$.
2. **Energy conservation**: each branch has the same p.d. V, so $I_1 = V/R_1$ and $I_2 = V/R_2$. For the whole combination, $I = V/R$.

3. Substitute: $V/R = V/R_1 + V/R_2$. Divide by V:

$$1/R = 1/R_1 + 1/R_2 + \ldots$$

68 Useful shortcuts

- Two resistors in parallel: **$R = R_1R_2 / (R_1 + R_2)$** ("product over sum").
- *n* equal resistors *R* in parallel: **R/n**.
- A parallel combination is **always less than the smallest resistor** in it. Adding another branch always **lowers** the total resistance (there is an extra path for charge).

68 Solving a network

1. Combine each parallel group into one resistance (invert at the end!).
2. Add the series parts to get the total resistance.
3. Total current $I = V/R_{total}$.
4. Work back: p.d. across each part = IR, then the current in each branch = V/R for that branch.

WORKED EXAMPLE: SERIES AND PARALLEL NETWORK

A 6.0 Ω resistor is in series with a parallel pair of 12 Ω and 4.0 Ω, across a 9.0 V supply (negligible internal resistance). Find the current in each resistor.
Parallel pair: R = (12 × 4.0)/(12 + 4.0) = 48/16 = **3.0 Ω**
Total: 6.0 + 3.0 = **9.0 Ω**, so I = 9.0/9.0 = **1.0 A** (this flows through the 6.0 Ω resistor)
p.d. across the 6.0 Ω = 1.0 × 6.0 = 6.0 V, so the parallel pair has 9.0 − 6.0 = 3.0 V across it
12 Ω: 3.0/12 = **0.25 A**; 4.0 Ω: 3.0/4.0 = **0.75 A**. Check: 0.25 + 0.75 = 1.0 A ✓

MARK-SCHEME LANGUAGE

- Series derivation (3 marks): "$V_T = V_1 + V_2$", "IR_T = $IR_1 + IR_2$", "I is the same, so cancel I".
- Parallel derivation (3 marks): "$I_T = I_1 + I_2$", "$V/R_T = V/R_1 + V/R_2$", "V is the same, so cancel V". The final equation is on the formulae sheet, so **quoting it alone scores nothing**: the marks are for the conservation steps.
- Calculations: "use of resistors in parallel formula" and "use of resistors in series formula" are separate marks. Ratio methods are accepted as well as current methods.

Often asked as: derive a formula (3 marks), find a total or an unknown resistance, "show that" a combination is about some value, which network expression is correct (MCQ).

COMMON MISTAKE

Forgetting to invert: 1/R = 11/3000 means R = 3000/11 = 273 Ω, not 3.7×10^{-3} Ω (Jan 2023 examiner report).
Leaving out a series resistor when adding up (using 15 Ω instead of 15 + 2 = 17 Ω, Oct 2023 examiner report).
Stating the conservation law without using it in the algebra.

SPEC 69 Electrical power and energy

WHAT THIS SPEC POINT ASKS

Use $P = VI$ and $W = VIt$, and derive and use $P = I^2R$ and $P = V^2/R$.

69 The equations

- Power is the rate of energy transfer. Energy per coulomb (V) × coulombs per second (I) = energy per second:

$$P = VI \qquad W = VIt$$

- P = power (watt, W = J s^{-1}); W = energy transferred (J); t = time (s).
- **Derive** the other two forms by substituting R = V/I:
  - put V = IR into P = VI → **$P = I^2R$**
  - put I = V/R into P = VI → **$P = V^2/R$**
- Choose the form that matches what you know: $P = I^2R$ when the **current** is fixed or known (e.g. series), $P = V^2/R$ when the **p.d.** is fixed (e.g. components in parallel across a supply).

69 Comparing powers

- Same current (series): bigger R → **more** power ($P = I^2R$).
- Same p.d. (parallel across a fixed supply): bigger R → **less** power ($P = V^2/R$). So in a house, a lower-resistance appliance is the more powerful one.
- For a fixed resistor, doubling V or I **quadruples** P.
- In every "does it go up or down?" answer, say **which quantity stays constant** before you use the equation.

WORKED EXAMPLE: A HEATER

A heater is rated 2.0 kW at 230 V.
Current: I = P/V = 2000/230 = **8.7 A**
Resistance: $R = V^2/P = 230^2/2000$ = **26 Ω**
Energy in 15 minutes: W = Pt = 2000 × 900 = **1.8×10^6 J**. In kilowatt-hours: 2.0 kW × 0.25 h = **0.50 kWh** (1 kWh = 3.6×10^6 J).

WORKED EXAMPLE: IS THE STUDENT RIGHT?

An 8.0 Ω heater runs from a 12 V supply (negligible internal resistance). A student says that adding a second 8.0 Ω heater in parallel will halve the total power, "because the current is shared".
One heater: $P = V^2/R = 12^2/8.0$ = **18 W**
Two in parallel: R = 8.0/2 = 4.0 Ω, $P = 12^2/4.0$ = **36 W**
The total power **doubles** (each heater still has 12 V across it), so **the student is wrong**.
(Two in **series** would give R = 16 Ω and P = 9.0 W: that is the arrangement that halves the power.)

MARK-SCHEME LANGUAGE

- "Use of P = VI" / "use of $P = I^2R$" / "use of $P = V^2/R$"; for comparisons, "use of a suitable power equation to conclude that less power is dissipated".
- A conclusion mark is lost if the answer also contains a wrong statement about one of the variables (WPH12 Jun 2019).
- Derive $P = V^2/R$ (2 marks): "P = VI and V = IR (or I = V/R)", then substitute.

Often asked as: power or energy of a component, derive $P = V^2/R$, "a student suggests…" claims about adding components, power from a point on an I–V graph, energy costs.

COMMON MISTAKE

Using $P = I^2R$ with a current that changes, or $P = V^2/R$ with a p.d. that changes, without saying what is constant.
Giving power when the question asks for energy (Oct 2019 examiner report).

BEYOND THE SPEC, BUT EXAMINED

- Kilowatt-hour conversions: 1 kWh = 1000 W × 3600 s = 3.6×10^6 J (WPH12 Jan 2021).
- Maximum power from a fixed supply happens when the total resistance is lowest, so compare the numbers and conclude (WPH12 Jun 2021).
- Non-ohmic lamps: two lamps in series do not have exactly 2× the resistance of one lamp at full p.d., because each is cooler and so has less resistance (WPH12 Jun 2023).

SPEC 70

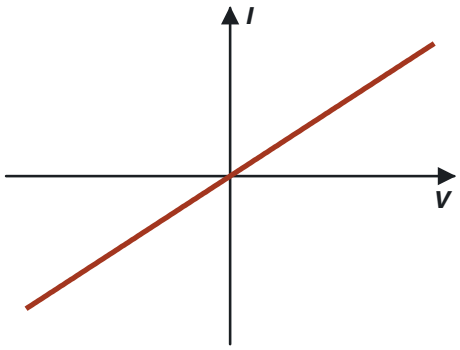
Current–p.d. graphs

WHAT THIS SPEC POINT ASKS

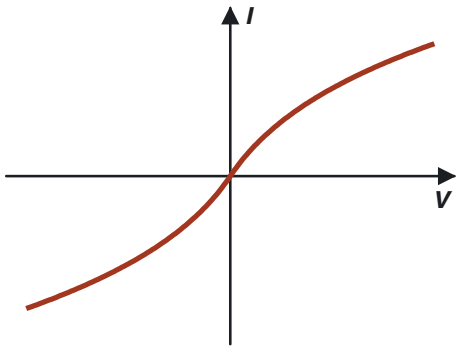
Sketch, recognise and interpret I–V graphs for ohmic conductors, filament bulbs, thermistors and diodes.

70 Getting the data

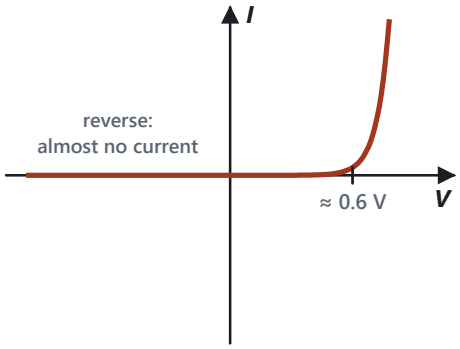
- Put the component in series with an **ammeter** and a way of varying the p.d. (a **variable resistor**, or a **potential divider** for a range down to 0 V). Connect a **voltmeter across the component**.
- Reverse the component (or the supply) to get **negative** values of V and I.
- For a diode, add a **protective resistor** in series so the forward current can't get too large.



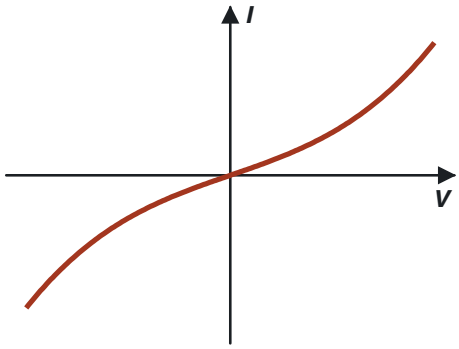
Ohmic conductor
straight line through the origin: R constant



Filament lamp
curve flattens: R rises as the filament heats



Semiconductor diode
conducts forwards above about 0.6 V



NTC thermistor
curve steepens: R falls as it heats

Current (y-axis) against p.d. (x-axis). The lamp, resistor and thermistor graphs are symmetrical through the origin: reversing the p.d. just reverses the current.

70 What each shape means

Component	Shape	Why
Ohmic conductor (resistor or metal wire at constant temperature)	straight line through the origin	R constant: I ∝ V
Filament lamp	curve that gets **less steep** at higher p.d., symmetrical	more current → filament hotter → ions vibrate more → more collisions → **R rises** (79)
NTC thermistor	curve that gets **steeper** at higher p.d., symmetrical	more current → it warms up → **more conduction electrons** → **R falls** (79)
Diode	almost zero current in reverse; current rises steeply above about **0.6 V** (silicon) forward	conducts in one direction only; very high R in reverse

- **Resistance at a point** = V/I at that point (66). On the lamp graph, V/I rises along the curve; on the thermistor graph it falls.
- An **LED** behaves like a diode, but with a higher switch-on p.d. (about 1.6–2 V).

WORKED EXAMPLE: USING AN I–V GRAPH

The I–V graph of a lamp passes through (6.0 V, 0.40 A). Find the resistance and the power at this p.d.
R = V/I = 6.0/0.40 = **15 Ω**
P = VI = 6.0 × 0.40 = **2.4 W**
Read values to within half a small square and quote them.

MARK-SCHEME LANGUAGE

- Sketch of a filament lamp: "correct shape in the positive quadrant" and "correct symmetry in the negative quadrant" (one mark each).
- Filament lamp explanation: "resistance increases as p.d. increases" / "current increases more slowly"; "temperature increases"; "ions/lattice vibrate more"; "more collisions between electrons and the lattice".
- Thermistor explanation: "resistance decreases as p.d. increases"; "more conduction electrons"; "I = nqvA, so the current increases".
- References to the gradient being 1/R are ignored.

Often asked as: identify a graph (MCQ), sketch the lamp or diode graph, a 6-mark comparison of a lamp and a thermistor, read a value and find R or P, draw the circuit for an LED or diode.

COMMON MISTAKE

Swapping the lamp and the thermistor: they behave in opposite ways (Oct 2019 examiner report).
Drawing the circuit with no variable resistor (so only one reading is possible), or drawing a thermistor symbol by mistake (Oct 2024 examiner report).
Sketching only the positive quadrant when the question shows both.

BEYOND THE SPEC, BUT EXAMINED

LEDs: a circuit with the LED **forward biased**, an ammeter in series, a voltmeter across the LED and a way of varying the p.d.; resistance "almost infinite" below the switch-on p.d. (WPH12 Oct 2024).

SPEC 71

Resistivity

WHAT THIS SPEC POINT ASKS

Use R = ρl/A.

71 Resistance of a wire

- A longer wire has more resistance (charge carriers collide with more ions): **R ∝ l**.
- A thicker wire has less resistance (more carriers side by side): **R ∝ 1/A**.
- The material sets the constant of proportionality, its **resistivity ρ**:

$$R = \rho l / A$$

- R = resistance (Ω); ρ = resistivity (**Ω m**); l = length (m); A = cross-sectional area (**m²**).
- **Resistivity** is the resistance of a sample of the material of **unit length and unit cross-sectional area** (or: ρ = RA/l with the terms defined). It depends on the **material and its temperature**, not on the shape.
- Units: ρ = RA/l → Ω × m² / m = **Ω m** (ohm metre, not ohm per metre).

71 Areas you will need

- Round wire: **$A = \pi r^2 = \pi d^2/4$**. A micrometer measures the **diameter**: halve it before squaring.
- Strip, ribbon or film: **A = width × thickness**.
- Conversions: 1 mm = 10^{-3} m, so **1 mm² = 10^{-6} m²**; 1 cm² = 10^{-4} m².

Material	Typical resistivity / Ω m (room temperature)
copper	1.7×10^{-8}
aluminium	2.7×10^{-8}
constantan	about 5×10^{-7}
nichrome	about 1.1×10^{-6}
silicon (pure)	about 10^{3}
glass, plastics	10^{10} to 10^{16}

The huge range is explained in 73.

WORKED EXAMPLE: RESISTANCE OF A WIRE

A nichrome wire is 1.50 m long with diameter 0.32 mm. $\rho = 1.10 \times 10^{-6}$ Ω m. Find R.
$r = 0.16$ mm $= 1.6 \times 10^{-4}$ m; $A = \pi(1.6 \times 10^{-4})^2 = 8.04 \times 10^{-8}$ m²
$R = \rho l/A = 1.10 \times 10^{-6} \times 1.50 / 8.04 \times 10^{-8}$ = **20.5 Ω**

WORKED EXAMPLE: REARRANGING FOR LENGTH OR RESISTIVITY

(a) What length of copper wire ($\rho = 1.7 \times 10^{-8}$ Ω m, diameter 0.50 mm) has a resistance of 0.50 Ω?
$A = \pi(2.5 \times 10^{-4})^2 = 1.96 \times 10^{-7}$ m²
$l = RA/\rho = 0.50 \times 1.96 \times 10^{-7} / 1.7 \times 10^{-8}$ = **5.8 m**
(b) A conducting strip is 2.0 cm wide, 0.15 mm thick and 12 cm long, with R = 4.8 Ω. Find ρ.
$A = 0.020 \times 1.5 \times 10^{-4} = 3.0 \times 10^{-6}$ m²
$\rho = RA/l = 4.8 \times 3.0 \times 10^{-6} / 0.12$ = **1.2×10^{-4} Ω m**

WORKED EXAMPLE: STRETCHING A WIRE

A wire is stretched to twice its length. Its volume (and resistivity) don't change. What happens to R?
Volume $V = Al$ is constant, so if l doubles, A halves.
$R = \rho l/A = \rho l^2/V$, so **$R \propto l^2$**: doubling l makes R **four times** bigger.

MARK-SCHEME LANGUAGE

- "Use of $R = \rho l/A$" and "use of $A = \pi r^2$" are separate marks. If the diameter is used as the radius, the area mark can still be given when the substitution is dimensionally right, but the answer will be wrong.
- Definition: "resistance of a sample of unit cross-sectional area and unit length".
- In a "show that" you must give one more significant figure than the stem (show 23.5 Ω for "about 24 Ω").

Often asked as: find ρ, l, A or a diameter (2–3 marks, almost every paper), rectangular cross-sections, choosing between copper and aluminium cables, stretching a wire at constant volume.

COMMON MISTAKE

Not halving the diameter, or forgetting π or the square. Powers of ten with mm, mm^2 and "milli" (65 mΩ = 0.065 Ω).
Mixing up **resistance** (Ω, depends on shape) and **resistivity** (Ω m, a property of the material): a frequent cause of zero marks (Jan 2022 examiner report).
Missing or wrong unit for resistivity: it is Ω m (Jan 2025 examiner report). Area is in m^2, never m^{-2}.
For a stretched wire, saying "R against I^2 is a straight line" when the question wants "R is **proportional** to I^2" (Oct 2024 examiner report).

BEYOND THE SPEC, BUT EXAMINED

- Constant-volume stretching: $R = \rho l^2/V$, so $R \propto I^2$ (WPH12 Jun 2026).
- Using a table of resistivity against temperature to estimate a filament temperature (WPH12 Jun 2022).

SPEC 72

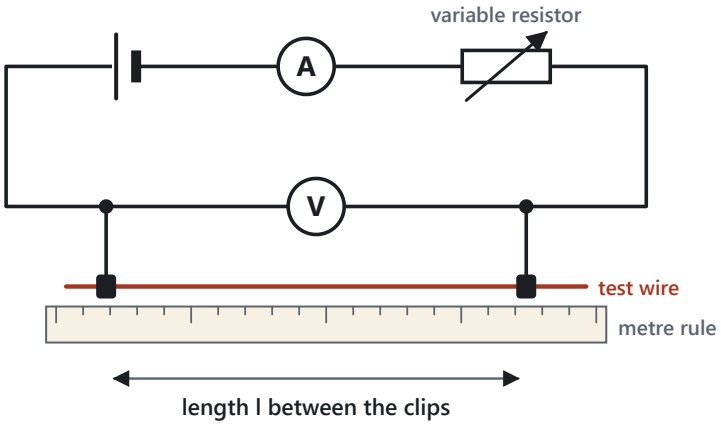
Core practical 7: resistivity of a material

WHAT THIS SPEC POINT ASKS

Measure the electrical resistivity of a material (a metal wire).

72 Principle

- Measure the resistance R of different lengths l of the same wire. Since R = (ρ/A) l, a graph of **R against l** is a straight line with **gradient ρ/A**. Measure the diameter to get A, then **ρ = gradient × A**.



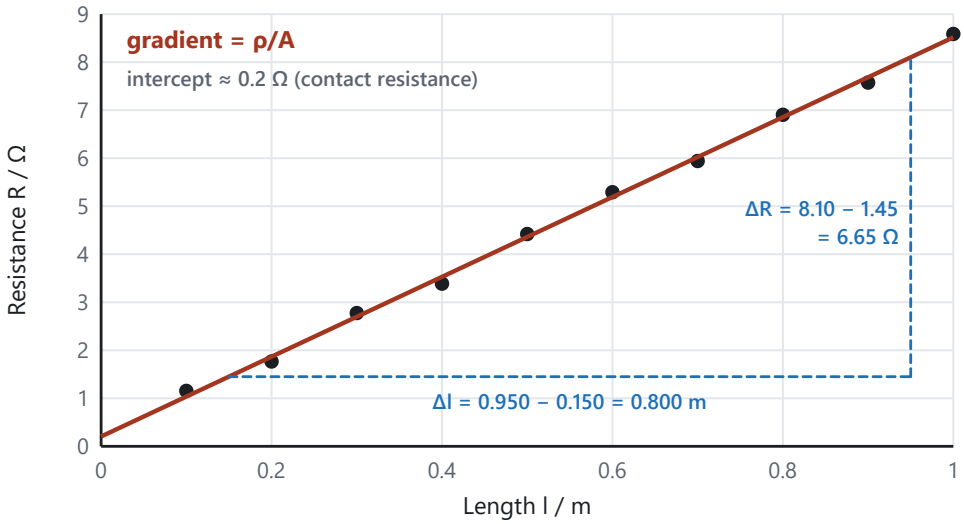
Ammeter in series, voltmeter across the length of wire between the crocodile clips. (An ohmmeter across the clips also works.)

CORE PRACTICAL 7: METHOD IN BRIEF

1. Measure the **diameter** of the wire with a **micrometer** at several places along it, turning it through 90° each time. Check the micrometer's **zero error** first. Take the mean.
2. Tape the wire along a **metre rule**. Connect it into the circuit with **crocodile clips**: cell, ammeter and variable resistor in series with the wire; voltmeter across the wire between the clips.
3. Set the clips 0.100 m apart. Close the switch, read V and I, then **open the switch** straight away.
4. Repeat for 8–10 lengths up to about 1 m. Repeat each reading and average. R = V/I for each length.
5. Plot R against l, draw the line of best fit and find the gradient.
6. ρ = gradient × A, where $A = \pi d^2/4$.

72 Variables and safety

- **Independent**: length l. **Dependent**: resistance R (from V and I). **Control**: the wire (same material and diameter) and its **temperature**, so keep the current **small** and the switch closed only briefly.
- **Safety**: the wire can get hot at low resistance (short lengths). Don't touch it while current flows, and use a low supply p.d.



A big gradient triangle (over half the line). The small intercept comes from contact resistance at the clips; the gradient is not affected by it.

WORKED EXAMPLE: P FROM THE GRADIENT

Diameter (mean of readings) d = 0.274 mm. From the graph, gradient = (8.10 − 1.45)/(0.950 − 0.150) = 6.65/0.800 = 8.31 Ω m^{-1}.

A = π(0.274 × 10^{-3})^2/4 = 5.90 × 10^{-8} m^2

ρ = gradient × A = 8.31 × 5.90 × 10^{-8} = **4.9 × 10^{-7} Ω m** (about right for constantan)

Uncertainty: d = 0.274 ± 0.002 mm is a 0.73% uncertainty, so A (which uses d^2) has **2 × 0.73 ≈ 1.5%** uncertainty. The diameter is usually the biggest source of uncertainty.

72 Errors and improvements

- **Heating** raises the resistance: small currents, switch off between readings.
- **Contact resistance** at the clips and the clips' width make l uncertain: the graph method puts this into the **intercept**, so the gradient (and ρ) is not affected. This is why a graph beats a single calculation.
- The wire may not be uniform: measure d at several points and orientations.
- Kinks in the wire make l too large: keep it straight and taut.

MARK-SCHEME LANGUAGE

- Method (3–4 marks): "micrometer (or digital calliper) for the diameter", "metre rule for the length", "voltmeter in parallel, ammeter in series" (or an ohmmeter), "use ρ = RA/l, or a graph gradient".
- Circuit (2 marks): supply, ammeter, variable resistor and wire in series; voltmeter across the wire.
- Graph (3 marks): "use of πr^2 or πd^2/4", "suitable axes", "the matching gradient gives ρ". Accepted pairs include R against l (gradient ρ/A), R against l/A (ρ), RA against l (ρ), l against R (A/ρ).
- Gradient: use a triangle over more than half the line; quote points read accurately.

Often asked as: describe the method, draw the circuit, choose the graph and say what the gradient is, process a gradient into ρ.

COMMON MISTAKE

Using a ruler or vernier for the diameter of a thin wire: you need a micrometer.
Using one pair of readings (V/I) instead of a gradient when the question gives a graph.

SPEC 73

I = nqvA and the range of resistivities

WHAT THIS SPEC POINT ASKS

Use I = nqvA to explain why resistivities of different materials cover such a huge range.

73 Deriving the transport equation

- Picture a conductor of cross-sectional area A with n charge carriers per cubic metre, each of charge q, drifting at average speed v.
- In a time t, carriers move a distance vt, so all the carriers in a volume Avt pass a point.
- Number passing = $nAvt$; charge passing = $nAvtq$.
- Current = charge / time:

$$I = nqvA$$

- n = **number density** of charge carriers (number of conduction electrons per m^3, unit **m^{-3}**); q = charge on each carrier (C; for electrons, e = 1.60 × 10^{-19} C); v = **drift velocity** (m s^{-1}); A = cross-sectional area (m^2).
- **Drift velocity** = the **average (mean) velocity of the charge carriers** along the conductor. The electrons move randomly at high speeds; the drift is a slow average movement superimposed on this.

73 Why resistivities range so widely

Type	n / m^{-3} (roughly)	Resistivity
metal	10^{28} to 10^{29}	very low
semiconductor	10^{15} to 10^{22} (rises with temperature and light)	in between
insulator	almost zero	enormous

- For the same current through the same area, **v ∝ 1/n**. Few carriers would need an enormous drift speed, which the p.d. can't produce, so the current is tiny and the resistivity is huge.
- So the **number of conduction electrons per unit volume** is the main reason resistivity ranges over about 24 powers of ten.
- Drift speeds in metals are tiny (about 10^{-4} m s^{-1}), yet a lamp lights at once: the electric field is set up round the whole circuit almost instantly, so **all** the electrons start drifting together.

WORKED EXAMPLE: DRIFT VELOCITY IN COPPER

A copper wire of diameter 0.80 mm carries 2.0 A. n = 8.5 × 10^{28} m^{-3}. Find the drift velocity.
A = π(0.40 × 10^{-3})^2 = 5.03 × 10^{-7} m^2
v = I/(nqA) = 2.0 / (8.5 × 10^{28} × 1.60 × 10^{-19} × 5.03 × 10^{-7}) = **2.9 × 10^{-4} m s^{-1}**
At this speed an electron takes about an hour to drift 1 m.

WORKED EXAMPLE: REASONING WITH I = NQVA

(a) A wire is cut to half its length and connected to the same p.d. What happens to the drift velocity?
R = ρl/A halves → I = V/R doubles → n, q and A are unchanged, so by I = nqvA **v doubles**.
(b) Two wires of the same metal are in series; wire Z has a quarter of the cross-sectional area of wire W.
Same current (series), same n and q, so v ∝ 1/A: **v in Z is 4 times v in W**.

MARK-SCHEME LANGUAGE

- "Use of I = nqvA (with a dimensionally correct A)". A minus sign on v is ignored.
- Drift velocity: "average/mean velocity of the (free) electrons" (accept "speed", "charge carriers").
- "Copper has a high concentration (n) of charge carriers, so v is small."
- Comparisons: say what changes and what stays constant (n and q for the same metal; I in series), then conclude.

Often asked as: calculate v (almost every paper), compare drift speeds in two wires, what happens when a wire's length or area changes, the meaning of n (MCQ).

COMMON MISTAKE

Using the electron's **mass** instead of its charge for q, or forgetting q altogether (Jan 2022, Jun 2019 examiner reports).
Using the diameter as the radius, or forgetting π or the square (Jan 2021 examiner report).
"Length isn't in I = nqvA, so changing it has no effect." It changes R, so it changes I, so it changes v (Jan 2022 examiner report).

BEYOND THE SPEC, BUT EXAMINED

Comparing two metals with the same current and area: the one with the larger n has the smaller drift speed (WPH12 Oct 2024). Comparing copper and aluminium cables needs both R and n, and sometimes you can't decide without more data (WPH12 Jan 2024).

SPEC 74

Potential along a uniform wire

WHAT THIS SPEC POINT ASKS

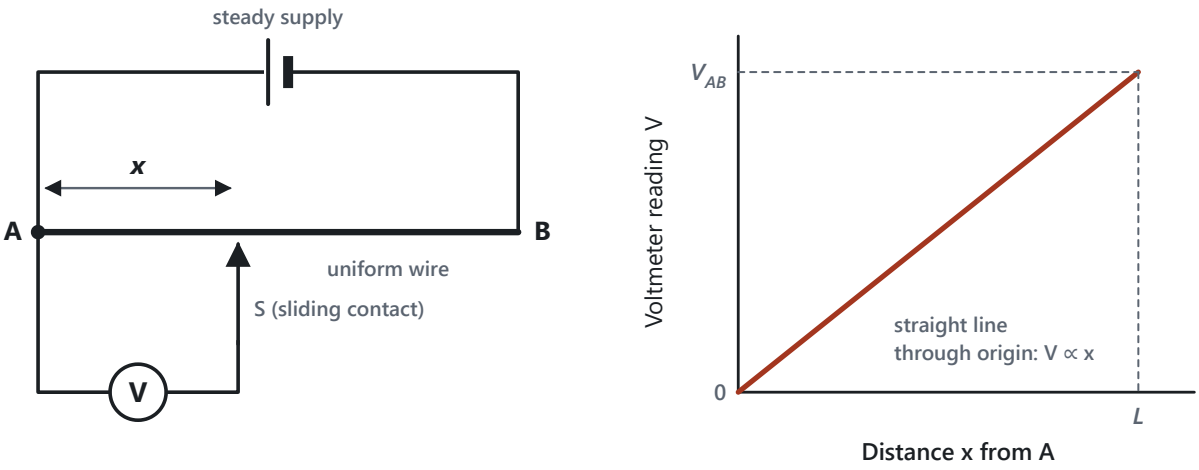
Understand how the potential varies with distance along a uniform wire carrying a current.

74 p.d. is proportional to length

- A **uniform** wire has the same material and the same cross-sectional area all along it. With a **steady current** I, the resistance of a length x is R = ρx/A, so the p.d. across that length is

$$V = IR = (I\rho/A)\,x \quad \text{so} \quad V \propto x$$

- The potential **falls steadily** (linearly) from the end connected to + to the end connected to −.
- A graph of p.d. against distance from one end is a **straight line through the origin**. Its gradient, the **potential gradient** (V m^{-1}), is constant: V_{AB}/L, which also equals Iρ/A. Here A is in m^2 and ρ in Ω m.
- With a sliding contact S on the wire, the wire acts as a **potential divider**: AS : SB = V_{AS} : V_{SB}.



The voltmeter reading rises in proportion to the distance x from A.

WORKED EXAMPLE: SLIDE-WIRE READINGS

(a) A uniform wire 1.00 m long has 2.0 V across it. A voltmeter is connected between one end and a contact 35.0 cm along.
V = 2.0 × (0.350/1.00) = **0.70 V**
(b) A cell of e.m.f. 1.50 V is connected across an 80.0 cm wire. Because the cell has internal resistance, the p.d. across the wire is only 1.44 V. Reading with the contact 25.0 cm from the end?
V = 1.44 × (25.0/80.0) = **0.45 V** (use the p.d. across the wire, not the e.m.f.)
(c) A graph of V against x for a wire carrying 0.50 A has gradient 3.6 V m^{-1}. A = 1.5 × 10^{-7} m^2. Find ρ.
gradient = Iρ/A, so ρ = gradient × A / I = 3.6 × 1.5 × 10^{-7} / 0.50 = **1.1 × 10^{-6} Ω m**

MARK-SCHEME LANGUAGE

- "p.d. is (directly) proportional to length"; V = V_{total} × x/l.
- Slide wire MCQs: "the ratio AS : SB equals R_1 : R_2".
- For a V against I graph: "gradient = ρI/A".

Often asked as: slide-wire MCQs, a voltmeter reading at part of a wire, ρ from a graph of V against length.

COMMON MISTAKE

Using the cell's e.m.f. when the wire actually has the (smaller) terminal p.d. across it (Oct 2023 examiner report).
Giving the full p.d. or 0 V as the reading at an intermediate point. Find the fraction of the length first.

SPEC 75

Potential dividers

WHAT THIS SPEC POINT ASKS

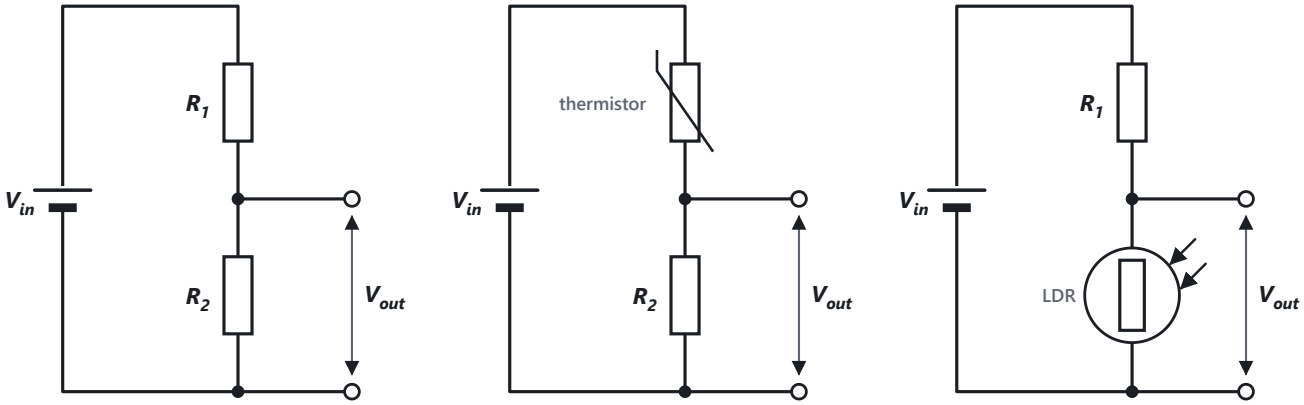
Understand how a potential divider works and calculate p.d.s and resistances in one.

75 How it works

- Two resistors in series across a supply V_{in} carry the **same current**, so the p.d. across each is **proportional to its resistance**. The supply p.d. is shared in the ratio of the resistances:

$$V_1 / V_2 = R_1 / R_2 \qquad V_{out} = V_{in} \times R_2 / (R_1 + R_2)$$

- V_{out} is the p.d. across R_2 (the resistor the output is taken across). The **bigger resistor takes the bigger share** of the p.d.
- Uses: providing a **fixed lower p.d.**, a **variable p.d.** (a potentiometer, see 76) and **sensor circuits** (thermistor or LDR, see 76).



(a) Fixed resistors
output p.d. is the share across R_2

(b) Thermistor on top
hotter → thermistor R falls → output rises

(c) LDR at the bottom
brighter → LDR R falls → output falls

The output is always the p.d. across the lower component here. Which component is a sensor, and where it sits, decides whether the output rises or falls.

75 Two ways to calculate

- **Ratio method**: $V_{out} = V_{in} \times R_2/(R_1 + R_2)$.
- **Current method**: $I = V_{in}/(R_1 + R_2)$, then $V_{out} = IR_2$. Use this when there are more than two resistors.

75 Loading a divider

- If you connect a **load** (or a voltmeter that is not ideal) across R_2, it is in **parallel** with R_2, so the resistance of that part **falls**, its share of the p.d. **falls**, and V_{out} is **lower** than calculated.
- To avoid this, the load's resistance should be much bigger than R_2.

WORKED EXAMPLE: A DIVIDER, THEN LOADING IT

A 9.0 V supply (negligible internal resistance) is across R_1 = 47 kΩ and R_2 = 22 kΩ in series.
(a) $V_{out} = 9.0 \times 22/(47 + 22) =$ **2.9 V**
(b) A voltmeter of resistance 100 kΩ is connected across R_2. What does it read?
R_2 in parallel with the voltmeter: $(22 \times 100)/(22 + 100) = 18.0$ kΩ
Reading $= 9.0 \times 18.0/(47 + 18.0) =$ **2.5 V**, lower than 2.9 V.
(c) To get 3.0 V from a 12 V supply with R_2 = 1.0 kΩ: R_1 must take 9.0 V, three times as much as R_2, so **R_1 = 3.0 kΩ**.

MARK-SCHEME LANGUAGE

- "Uses ratio of resistances to p.d.s" is the standard first mark.
- Why a measured V_{out} is lower than calculated (one mark for each reason with its explanation): "the cell has internal resistance, so the terminal p.d. is less than the e.m.f."; "the wires/connections have resistance, so some p.d. is across them"; "the voltmeter has finite resistance, so it draws current and the parallel combination has less resistance than the resistor".

Often asked as: voltmeter reading across one resistor, a minimum or maximum output, an unknown resistance, why the measured value differs from the calculated one.

COMMON MISTAKE

Giving the p.d. across the **other** resistor (Oct 2020 examiner report: the common wrong answer). Check which component the output is taken across.
Explaining a low reading using only wire resistance and ignoring the voltmeter's own resistance.

SPEC 76

Potential dividers with sensors

WHAT THIS SPEC POINT ASKS

Analyse potential dividers where one resistance can vary, including thermistors and light-dependent resistors (LDRs).

76 Potentiometer

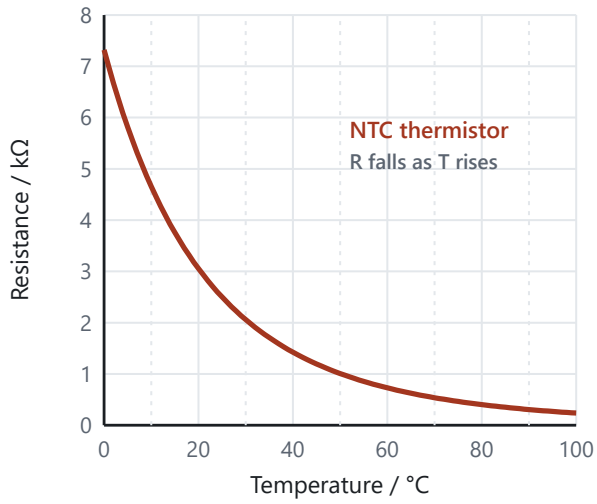
- A **potentiometer** is a resistor with a **sliding contact**: the slider splits it into two parts, R_1 and R_2, that form a divider. Moving the slider gives any output from **0 V up to V_{in}**.
- A variable resistor used in series with a component (as a rheostat) can't bring the p.d. across that component down to zero; a potentiometer can.

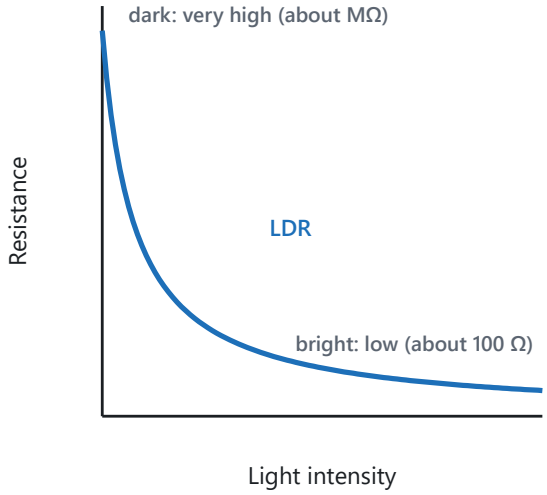
76 Sensor circuits

- A **thermistor** (NTC): resistance **falls as temperature rises** (79).
- An **LDR**: resistance **falls as light intensity rises** (80).
- Put the sensor in series with a fixed resistor. The sensor's share of the p.d. follows its share of the resistance.

Change	Sensor R	p.d. across sensor	p.d. across fixed resistor
thermistor gets hotter	falls	falls	rises
LDR gets brighter	falls	falls	rises
colder / darker	rises	rises	falls

- So **where you take the output from** decides the behaviour: output across the fixed resistor rises with temperature or light; output across the sensor falls.
- The total current also changes: lower sensor R → lower total R → bigger current.





Read sensor resistances off graphs like these. Left: the thermistor used in the worked example below (about 3.0 kΩ at 20 °C, 1.4 kΩ at 40 °C, 1.0 kΩ at 50 °C).

76 The four-step method for sensor questions

1. **Read** the sensor's resistance from the graph at the stated temperature or light level, and quote it.
2. **Calculate** the p.d. (ratio of resistances, or $I = V/R_{total}$ then $V = IR$).
3. **Compare** with the threshold in the question.
4. **Conclude** in words: "so the fan switches on" / "so the student is wrong".

WORKED EXAMPLE: A THERMISTOR-CONTROLLED FAN

In circuit (b) of the divider figure, the thermistor is in series with a 2.0 kΩ resistor across a 6.0 V supply (negligible internal resistance). The output is the p.d. across the 2.0 kΩ resistor. A fan switches on when the output reaches 3.5 V. Use the thermistor graph.
(a) At 20 °C, $R_{th} \approx 3.0$ kΩ: $V_{out} = 6.0 \times 2.0/(3.0 + 2.0) =$ **2.4 V**. That is less than 3.5 V, so the fan is **off**.
(b) At 50 °C, $R_{th} = 1.0$ kΩ: $V_{out} = 6.0 \times 2.0/(1.0 + 2.0) =$ **4.0 V**. That is more than 3.5 V, so the fan is **on**.
(c) Switch-on temperature: thermistor p.d. = 6.0 − 3.5 = 2.5 V, so R_{th} / 2.0 kΩ = 2.5/3.5, giving R_{th} = **1.4 kΩ**. From the graph, that is about **40 °C**.

WORKED EXAMPLE: AN LDR LIGHT SENSOR

An LDR is in series with a 4.7 kΩ resistor across 5.0 V, with the output across the LDR. In the dark R_{LDR} = 100 kΩ; in bright light R_{LDR} = 300 Ω.
Dark: $V_{out} = 5.0 \times 100/(100 + 4.7) =$ **4.8 V**
Bright: $V_{out} = 5.0 \times 0.30/(0.30 + 4.7) =$ **0.30 V**
The output is high in the dark, so it could switch on a street light.

EXAM TIP

For a 6-mark "explain what happens" question, write a chain with one link per sentence: temperature rises → more conduction electrons in the thermistor → its resistance falls → it takes a smaller share of the total resistance → smaller p.d. across it → larger p.d. across the fixed resistor → the voltmeter reading rises. Name a **specific** change ("temperature rises"), not just "the temperature changes".

MARK-SCHEME LANGUAGE

- Calculation route: read R from the graph, "ratio of resistances = ratio of p.d.s" (or the potential divider equation), compare and conclude.
- Model conclusion (WPH12 Oct 2023): "If the temperature goes below this level then the resistance of the thermistor increases, so the p.d. to the heater switch increases, so the switch does perform as required."
- 6-mark schemes often "allow converse statements" but do not allow contradictory statements about the current or p.d.

Often asked as: one sensor question in almost every paper: the p.d. or temperature at which something switches, a 6-mark reasoning chain, MCQs about what happens when light or temperature changes.

COMMON MISTAKE

Stopping at the number without explaining what it means for the circuit (Oct 2023 examiner report: candidates were expected to explain what happens below 10 °C).
Using the wrong p.d. in R = V/I, or not subtracting the fixed resistor from the total (Jan 2021 examiner report).
Describing the sensor well but never reaching the circuit: the circuit marks are the ones most often missed (Oct 2019 examiner report).

BEYOND THE SPEC, BUT EXAMINED

- If an LDR's resistance falls at a fixed e.m.f., the total power $P = V^2/R_{total}$ **rises** (WPH12 Jun 2023).
- A voltmeter's **resolution** as an argument: a p.d. smaller than the last digit can't be detected (WPH12 Jan 2021).
- Doubling the supply e.m.f. **more than** doubles a thermistor-circuit reading: the bigger current heats the thermistor, so its resistance falls too (WPH12 Jun 2021).

SPEC 77

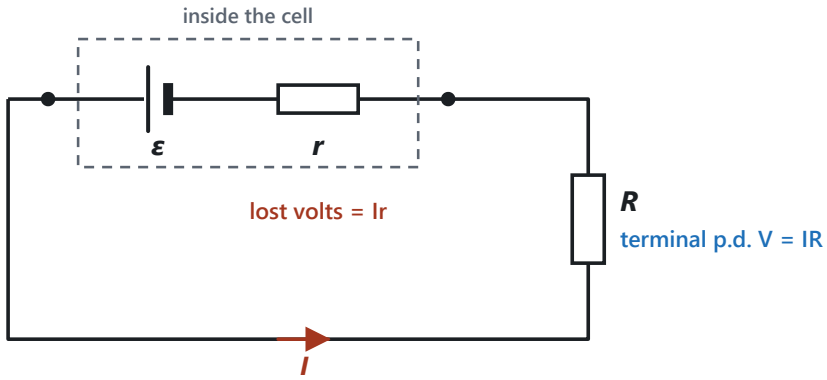
E.m.f. and internal resistance

WHAT THIS SPEC POINT ASKS

Define e.m.f., understand internal resistance, and tell e.m.f. apart from terminal p.d.

77 Definitions

- **Electromotive force (e.m.f.), ε**: the **energy transferred to electrical energy per unit charge** passing through the source (the work done by the source per coulomb). Unit: volt. It is **not a force**.
- **p.d.**: energy transferred **from** electrical energy to other forms per unit charge, in a component.
- **Internal resistance, r**: the resistance of the materials inside the source. Some energy per coulomb is dissipated as heat **inside** the cell.
- **Terminal p.d., V**: the p.d. across the cell's terminals, which equals the p.d. across the external circuit.
- **Lost volts**: the p.d. across the internal resistance, **Ir**.



ε = V + Ir = I(R + r)

Model a real cell as a perfect e.m.f. in series with a small resistor r.

77 The equation

- Energy conservation round the loop: e.m.f. = p.d. across R + p.d. across r:

$$\varepsilon = I(R + r) \qquad \varepsilon = V + Ir$$

- **No current** (open circuit, or a very high-resistance voltmeter alone across the cell): Ir = 0, so the **terminal p.d. = e.m.f.**
- Lower external R → bigger current → bigger lost volts → **terminal p.d. falls**.
- **Short circuit** (R = 0): the current is the largest possible, I = ε/r, and all the e.m.f. is across r.
- Applications: a car's headlights dim when the starter motor draws a huge current (big lost volts). High-voltage supplies are given a **very large** internal resistance so the current can never be dangerous.

WORKED EXAMPLE: TERMINAL P.D. AND LOST VOLTS

A cell has ε = 6.0 V and r = 0.80 Ω. It is connected to a 4.0 Ω resistor.
I = ε/(R + r) = 6.0/4.8 = **1.25 A**
Terminal p.d. V = IR = 1.25 × 4.0 = **5.0 V**; lost volts = Ir = 1.25 × 0.80 = **1.0 V** (5.0 + 1.0 = 6.0 ✓)
Power wasted in the cell = I^2r = 1.25^2 × 0.80 = **1.3 W**. Short-circuit current = 6.0/0.80 = **7.5 A**.

WORKED EXAMPLE: FINDING R

A high-resistance voltmeter reads 1.50 V across a cell on open circuit. With a 2.0 Ω resistor connected, it reads 1.20 V.
ε = 1.50 V. Current I = V/R = 1.20/2.0 = 0.60 A
Lost volts = 1.50 − 1.20 = 0.30 V, so r = 0.30/0.60 = **0.50 Ω**

WORKED EXAMPLE: TEST A CLAIM

A 12.0 V battery drives a 2.0 Ω heater. As the battery ages, r rises from 0.10 Ω to 0.50 Ω. A student claims the heater's power falls by **at least 30%**.
Before (r = 0.10 Ω): I = 12.0/2.1 = 5.71 A, P = I^2R = 5.71^2 × 2.0 = 65.3 W
After (r = 0.50 Ω): I = 12.0/2.5 = 4.80 A, P = 4.80^2 × 2.0 = 46.1 W
46.1/65.3 = 0.706, so the power falls by about **29%**. That is less than 30%, so **the claim is wrong**.

MARK-SCHEME LANGUAGE

- e.m.f.: "energy (supplied) to/per unit charge", "work done per unit charge", "work done moving unit charge round the whole circuit" or "terminal p.d. when the current is zero".
- Terminal p.d. lower than e.m.f. (3 marks): "p.d. is energy per unit charge"; "energy is dissipated in the battery (internal resistance)"; "so less energy per coulomb is available to the circuit". "Lost volts" is accepted.
- "There was no current in the battery, so no p.d. across the internal resistance."

Often asked as: define e.m.f., explain why the terminal p.d. drops, calculate r, the effect of changing R or of a larger current, "student claim" power comparisons, cells in series v parallel.

COMMON MISTAKE

"With nothing connected there is no e.m.f." The e.m.f. is still there; it's the current (and so the lost volts) that is zero (Jan 2024 examiner report).
Writing ε = V − Ir. It is **ε = V + Ir**.
Calling e.m.f. a force, or defining it as "energy per unit current".

BEYOND THE SPEC, BUT EXAMINED

- Combining cells: in **series** the e.m.f.s add and the internal resistances add; identical cells in **parallel** keep the same e.m.f., but the internal resistance of the combination is lower (r/2 for two), so fewer volts are lost for a given current. Work out the current and the lost volts for each arrangement before comparing terminal p.d.s (WPH12 Jun 2019).
- Larger load power (e.g. a motor going uphill) → larger current → larger p.d. across r → more energy wasted as heat inside the battery (WPH12 Jun 2025).

SPEC 78

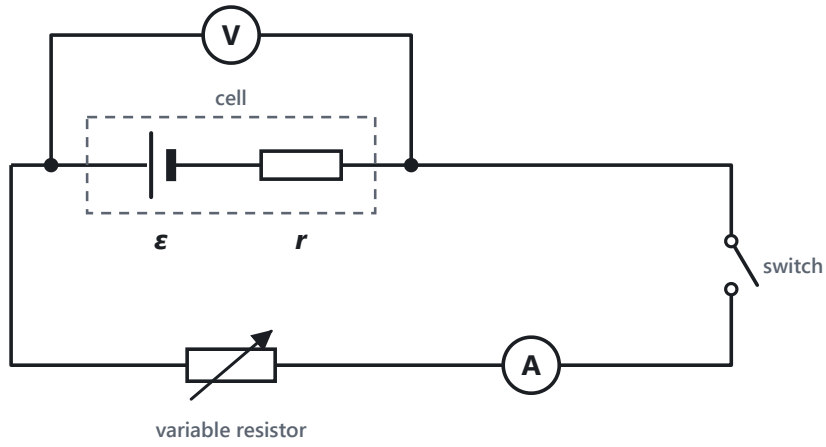
Core practical 8: e.m.f. and internal resistance of a cell

WHAT THIS SPEC POINT ASKS

Measure the e.m.f. and internal resistance of a cell.

78 Principle

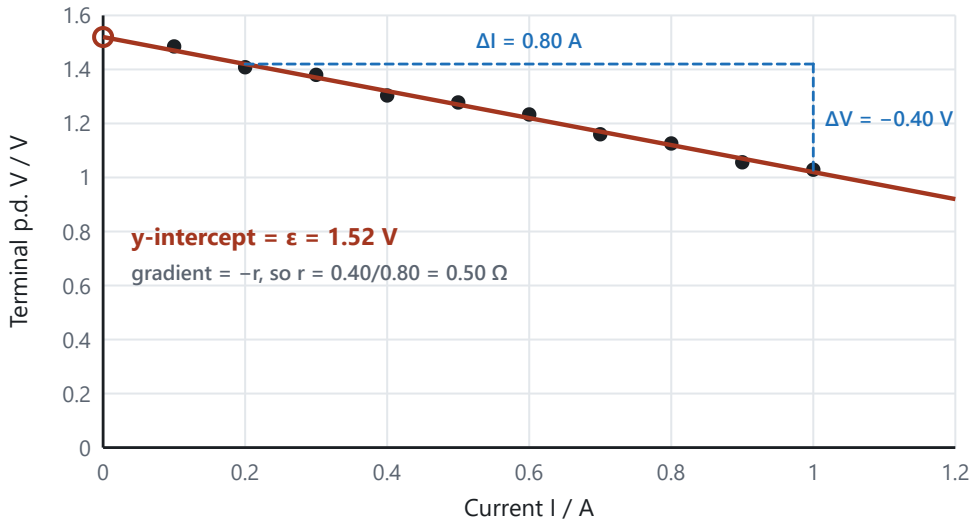
- Rearrange ε = V + Ir as **V = −rI + ε**. Compare with y = mx + c: a graph of **terminal p.d. V (y) against current I (x)** is a straight line with **gradient −r** and **y-intercept ε**.



Voltmeter across the cell's terminals; ammeter and variable resistor in series.

CORE PRACTICAL 8: METHOD IN BRIEF

1. Connect the cell in series with an **ammeter**, a **variable resistor** and a **switch**. Connect a **voltmeter across the cell's terminals**.
2. Set the variable resistor to its maximum, close the switch, record V and I, then **open the switch**.
3. Lower the resistance in steps to get 8–10 pairs of readings over as wide a range of current as possible. Keep the switch open between readings.
4. Repeat and average. Plot V against I and draw the line of best fit.
5. ε = the y-intercept; r = −gradient.



Terminal p.d. falls as the current rises because the lost volts (Ir) grow.

WORKED EXAMPLE: E AND R FROM THE GRAPH

Two points on the line of best fit: (0.20 A, 1.42 V) and (1.00 A, 1.02 V).

gradient = (1.02 − 1.42)/(1.00 − 0.20) = −0.40/0.80 = −0.50 V A^{-1}, so **r = 0.50 Ω**

ε = intercept = 1.42 + 0.50 × 0.20 = **1.52 V** (or read it off the y-axis at I = 0)

78 Other graphs that work

Plot	Gradient	Intercept
V against I	$-r$	ε (on the V axis)
I against V	$-1/r$	
$(\varepsilon - V)$ against I	r	0
R against 1/I (from $R = \varepsilon/I - r$)	ε	$-r$ (on the R axis)

78 Errors, improvements and safety

- The cell **warms up** and **runs down** if current flows for long, so ε and r drift: open the switch between readings.
- Use a wide range of currents so the gradient is well defined. Use digital meters with suitable resolution.
- **Safety**: never short-circuit the cell (large current, hot wires and cell). Keep currents modest.

MARK-SCHEME LANGUAGE

- Circuit (2–3 marks): "voltmeter in parallel with the cell", "ammeter in series", "variable resistor". The voltmeter may be across the variable resistor if nothing else is in the circuit.
- Equation and graph (4 marks): "$\varepsilon = V + Ir$ (from conservation of energy)", "compare with $y = mx + c$", "e.m.f. is the y-intercept", "internal resistance is the negative of the gradient".
- "If no best-fit line has been drawn, only some marks are available." Units are needed on ε and r.

Often asked as: draw the circuit, explain how the graph gives ε and r, read ε and r off a graph, an R against 1/I graph, predict the new line if ε and r both double (gradient and intercept both double).

COMMON MISTAKE

No variable resistor in the circuit (only one pair of readings possible), the voltmeter in the wrong place, or a thermistor drawn instead (Oct 2022, Jan 2024 examiner reports).
Dividing one V by one I to get r instead of using the gradient (Jan 2024 examiner report: max 1 mark).
Contradictions such as "the gradient is −r, so r is the inverse of the gradient", or "the gradient is R because R = V/I" (Oct 2022 examiner report).
Leaving the unit off ε (Jun 2023 examiner report).

SPEC 79 Resistance and temperature

WHAT THIS SPEC POINT ASKS

Model how resistance changes with temperature using lattice vibrations and the number of conduction electrons, for metals and NTC thermistors.

79 The two factors

- The current depends on **how many** charge carriers there are (n) and **how freely** they drift (collisions with the vibrating positive ions of the lattice). $I = nqvA$ ties them together.
- Raising the temperature does two things:
  - the ions of the **lattice vibrate more** (bigger amplitude), so electrons **collide with the lattice more often**, which reduces the drift velocity: R **up**;

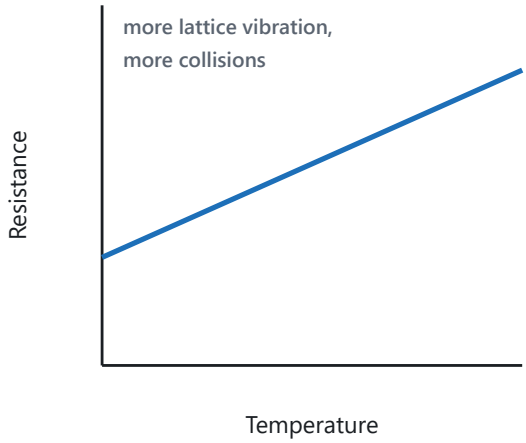
- in a semiconductor, more electrons gain enough energy to become **conduction electrons**, so n increases: R **down**.

79 Metals

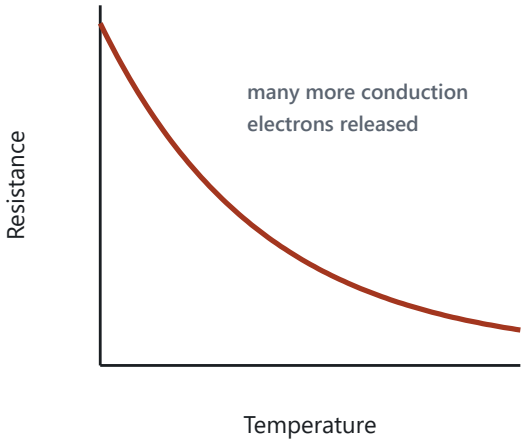
- In a metal, n is already huge and **does not change noticeably** with temperature.
- Temperature rises → lattice ions vibrate with greater amplitude → **more frequent collisions** between conduction electrons and the lattice → lower drift velocity for the same p.d. → **resistance increases**.
- This is why a filament lamp's I–V graph flattens (70).

79 NTC thermistors

- A thermistor is made of a **semiconductor**. **NTC** = **negative temperature coefficient**: its resistance **falls** as temperature rises.
- Temperature rises → electrons gain energy and are released → **many more conduction electrons** (n rises a lot) → **resistance decreases**.
- The lattice vibrations also increase, but the effect of the extra conduction electrons is **much bigger**, so overall R falls.



Metal: R rises with temperature



NTC thermistor: R falls

Same two effects, different winners: in a metal only the extra collisions matter; in a thermistor the extra conduction electrons dominate.

WORKED EXAMPLE: MODEL ANSWERS

Explain why the resistance of a metal wire increases as its temperature rises. (3 marks)

As the temperature rises, the lattice ions vibrate with a larger amplitude. So conduction electrons collide with the lattice more frequently. So the resistance increases.

Explain why the resistance of an NTC thermistor decreases as its temperature rises. (2 marks)

As the temperature rises, more electrons gain enough energy to become conduction electrons (n increases). So the resistance decreases (by I = nqvA, more current for the same p.d.).

MARK-SCHEME LANGUAGE

- Metal: "the (amplitude of) lattice vibrations increase"; "an increase in the frequency/rate of collisions of electrons with the lattice"; "resistance increases". Allow atoms/ions for lattice. The "resistance increases" mark depends on one of the first two being scored.
- Thermistor: "more conduction (free) electrons" or "more charge carriers", so "resistance decreases". Allow the converse.
- "Thermistors are semiconductors."

Often asked as: 3-mark metal explanation, 2-mark thermistor explanation, MCQs on what happens to lattice vibrations and electron numbers, combined circuits with a thermistor and a metal wire.

COMMON MISTAKE

Writing about the thermistor when the question is about the metal (more than half of candidates scored 0 on this in Jan 2025, examiner report).
Vague words: "particles", "atoms collide with each other", "electrons collide with each other". Say **electrons collide with the lattice (ions)**.
Saying "a change in temperature causes a change in electrons" when a **specific** change is needed (Oct 2023 examiner report).
For a thermistor, thinking lattice vibrations decrease when it gets hotter. They increase; the increase in n simply outweighs them (Jan 2022 examiner report on an MCQ).

BEYOND THE SPEC, BUT EXAMINED

- In a circuit with a thermistor and a metal-wire resistor, if the current falls as the temperature rises, the wire's increase in resistance must be greater than the thermistor's decrease (WPH12 Jan 2024).
- A filament's temperature rise lowers its power at fixed p.d., by $P = V^2/R$ (WPH12 Jun 2021).

SPEC 80

Resistance and illumination

WHAT THIS SPEC POINT ASKS

Model how resistance changes with light intensity in terms of the number of conduction electrons, and apply it to LDRs.

80 How an LDR works

- An **LDR** (light-dependent resistor) is made of a **semiconductor**.
- Light is a stream of **photons**. When a photon is absorbed, its energy can release an electron so it becomes a **conduction electron**.
- **Brighter light** → **more photons per second** absorbed → **more conduction electrons** (n rises) → **resistance falls** (I = nqvA: more current for the same p.d.).
- Typical values: in the **dark**, about a **megaohm**; in **bright light**, about **100 Ω**.
- At a **fixed** light level and temperature, an LDR's I–V graph is a **straight line through the origin**. Brighter light gives a **steeper** line (lower R).
- Uses: automatic street lights, garden lights, screen brightness control, alarms. In a divider (76), the output across the fixed resistor rises as light increases.

WORKED EXAMPLE: A 6-MARK STYLE CHAIN

An LDR is connected to a cell that has internal resistance, with a voltmeter across the cell's terminals. Explain what happens to the voltmeter reading when the light gets brighter.

The light intensity increases, so more photons per second are incident on the LDR.
So more electrons are released: the number of conduction electrons increases.
So the resistance of the LDR decreases, and the total resistance of the circuit decreases.
So the current increases ($I = \varepsilon/(R + r)$).
The internal resistance is constant, so the p.d. across it (Ir, the lost volts) increases.
So the terminal p.d. ($V = \varepsilon - Ir$) decreases: **the voltmeter reading falls**.

MARK-SCHEME LANGUAGE

- "Increase in light intensity: more photons per second incident on the LDR, so the number of (conduction) electrons increases, so the resistance decreases."
- Divider follow-on: "the LDR has a greater/smaller p.d.", "the p.d. across the fixed resistor decreases/increases", then a power equation for the power change.
- In some schemes the converse is **not** allowed for the first two marks: answer the direction of change the question asks about.

Often asked as: explain the LDR's response (2–4 marks), 6-mark chains through a divider or a cell's internal resistance, MCQs on readings when light changes.

COMMON MISTAKE

Mixing up the LDR and the thermistor: an LDR responds to **light** (photons), a thermistor to **temperature**.
"Light gives the LDR more energy so it has more resistance." More energy frees more electrons, so the resistance **falls**.
Leaving out photons: the marks want "more photons (per second)".

BEYOND THE SPEC, BUT EXAMINED

Linking the photon model (57–62) to the LDR: each absorbed photon can free one electron, so a higher **intensity** (more photons per second) gives more conduction electrons (WPH12 Jun 2026).